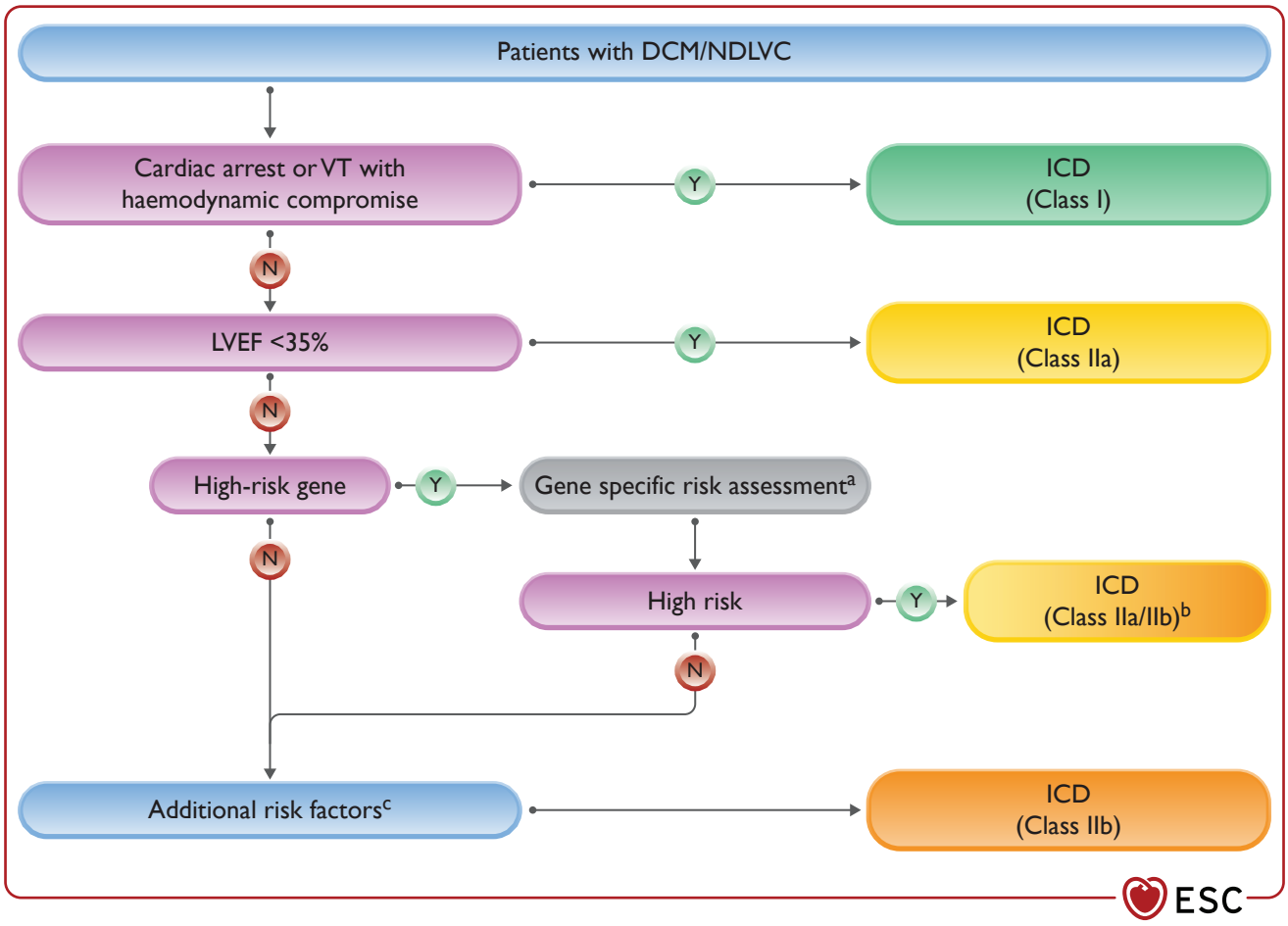


Figure 17 Implantation of implantable cardioverter defibrillators in patients with dilated cardiomyopathy or non-dilated left ventricular cardiomyopathy flowchart. CMR, cardiac magnetic resonance; DCM, dilated cardiomyopathy; ICD, implantable cardioverter defibrillator; LGE, late gadolinium enhancement; LVEF, left ventricular ejection fraction; NDLVC, non-dilated left ventricular cardiomyopathy; VE, ventricular ectopic beats; VT, ventricular fibrillation. [a]See *Table 21*. [b]Strength of recommendation depends on gene and context. [c]Additional risk factors include syncope, LGE presence on CMR.

p.Arg14del variant, variant-specific [https://plnriskcalculator.shinyapps.io/final_shiny])[542] risk-prediction scores have been developed that consider genotype characteristics and additional phenotypic features. Where such scores are available, they should be used to guide primary prevention ICD implantation (*Figure 17*). As discussed in *Section 7.1.5*, the Task Force acknowledges the challenges associated with defining universal thresholds for acceptable risk across different cardiomyopathy phenotypes, but is of the opinion that a similar approach to that taken in risk stratification for HCM is reasonable. Although the *2022 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death* have suggested a higher threshold of 10% risk at 5 years to guide primary prevention ICD implantation in patients with DCM and *LMNA* variants,[3] this Task Force recommends shared decision-making based on real-world data as well as individual preferences, beliefs, circumstances, and values. Gaps in evidence should be shared with patients, and competing risks related to the disease (heart failure, stroke), and to age and comorbidity, as well as device-related complications, should be discussed. In support of this, a recent study validating the LMNA-risk VTA calculator overestimated arrhythmic risk when using ≥7% predicted 5-year risk as threshold (specificity 26%, C-statistic 0.85), despite a high sensitivity.[871]

Importantly, there are also data to suggest that other genotypes (e.g. *TTN* truncating variants) are associated with recovery of LVEF with standard heart failure criteria from the *2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure*.[185,867,872]

In patients without a high-risk genotype and LVEF >35%, the presence and extent of myocardial scarring determined by LGE on CMR imaging can be helpful in risk stratification in patients with DCM.[873,874] Late gadolinium enhancement is observed in 25–35% of patients with DCM and its presence is a strong risk marker for all-cause mortality and ventricular arrhythmias, both in retrospective and prospective studies. A recent prospective study of 1020 DCM patients with a median follow-up of 5.2 years showed that myocardial scar had a strong and incremental prognostic value for predicting SCD, while LVEF ≤35% was not associated with SCD.[138] In another study, a risk calculator was developed that among others, incorporated the presence of LGE on CMR imaging[540], although this has not yet been externally validated. There are at least two ongoing trials of ICD therapy according to the presence of scar on CMR imaging, including DCM patients (NCT04558723 and NCT03993730), but the Task Force's opinion is that the existing level of evidence can support using LGE to guide ICD implantation in subgroups of patients with DCM (*Figure 17*). Additional risk factors, such as syncope or the presence of NSVT and burden of ventricular ectopy (VE), may also help guide ICD implantation. There are no data currently to support a specific threshold for VE burden, and this will depend on the underlying genotype and other

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clinical factors.[542,867,872] In patients with unexplained syncope, programmed electrical stimulation (PES) may provide additional information on the underlying cause.[875] There are no definitive data supporting the routine use of PES for primary prevention risk stratification in patients with DCM, but this may be beneficial in patients with DCM and myotonic dystrophy with an independent indication to electrophysiological study to assess conduction disturbances,[876] although the clinical value of this approach has not been consistently demonstrated.[877]

Recommendation Table 24 — Recommendations for an implantable cardioverter defibrillator in patients with dilated cardiomyopathy

Recommendations	Class[a]	Level[b]
Secondary prevention		
An ICD is recommended to reduce the risk of sudden death and all-cause mortality in patients with DCM who have survived a cardiac arrest or have recovered from a ventricular arrhythmia causing haemodynamic instability.[530,531,884]	I	B
Primary prevention		
An ICD should be considered to reduce the risk of sudden death and all-cause mortality in patients with DCM, symptomatic heart failure, and LVEF ≤35% despite >3 months of OMT.[861,885]	IIa	A
The patient's genotype should be considered in the estimation of SCD risk in DCM.[185,186,869,886]	IIa	B
An ICD should be considered in patients with DCM with a genotype associated with high SCD risk and LVEF >35% in the presence of additional risk factors (see *Table 21*).[541,542,867,869,873,878,881,886]	IIa	C
An ICD may be considered in selected patients with DCM with a genotype associated with high SCD risk and LVEF >35% without additional risk factors (see *Table 21*).[869,873,881,886]	IIb	C
An ICD may be considered in patients with DCM without a genotype associated with high SCD risk and LVEF >35% in the presence of additional risk factors.[c,138,873,874]	IIb	C

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CMR, cardiac magnetic resonance; DCM, dilated cardiomyopathy; ICD, implantable cardioverter defibrillator; LGE, late gadolinium enhancement; LVEF, left ventricular ejection fraction; OMT, optimal medical therapy; SCD, sudden cardiac death.
[a]Class of recommendation.
[b]Level of evidence.
[c]Additional risk factors include syncope, LGE presence on CMR.

7.3. Non-dilated left ventricular cardiomyopathy

7.3.1. Diagnosis

7.3.1.1. Index case

The non-dilated LV cardiomyopathy phenotype is defined by the presence of non-ischaemic LV scarring or fatty replacement in the absence of LV dilatation, with or without global or regional wall motion abnormalities, or isolated global LV hypokinesia without scarring (as assessed by the presence of LGE on CMR) that is unexplained solely by abnormal loading conditions (hypertension, valve disease) or CAD. Global LV systolic dysfunction is defined by abnormal LVEF (i.e. <50%).[9]

Table 21 High-risk genotypes and associated predictors of sudden cardiac death

Gene	Annual SCD rate	Predictors of SCD
LMNA[185,186,438,541,865,878,879]	5–10%	Estimated 5-year risk of life-threatening arrhythmia using *LMNA* risk score (https://lmna-risk-vta.fr)
FLNC-truncating variants[866,867,880]	5–10%	LGE on CMR LVEF < 45%
TMEM43[868,881]	5–10%	Male Female and any of the following: LVEF <45%, NSVT, LGE on CMR, >200 VE on 24h Holter ECG
PLN[542,882,883]	3–5%	Estimated 5-year risk of life-threatening arrhythmia using *PLN* risk score (https://plnriskcalculator.shinyapps.io/final_shiny) LVEF < 45% LGE on CMR NSVT
DSP[185,186]	3–5%	LGE on CMR LVEF < 45%
RBM20[869]	3–5%	LGE on CMR LVEF < 45%

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CMR, cardiac magnetic resonance; DSP, desmoplakin; ECG, electrocardiogram; FLNC, filamin C; LGE, late gadolinium enhancement; LMNA, lamin A/C; LVEF, left ventricular ejection fraction; NSVT, non-sustained ventricular tachycardia; PLN, phospholamban; RMB, RNA binding motif protein; SCD, sudden cardiac death; VE, ventricular ectopic beats.

7.3.1.2. Relatives

Clinical testing in relatives may reveal non-diagnostic abnormalities. In this context, the presence of LV systolic global or regional dysfunction, or additional electrocardiographic abnormalities (e.g. repolarization abnormalities, low QRS voltages, frequent ventricular extrasystoles [>500 per 24 h] or NSVT) in a first-degree relative of an individual with NDLVC (or a first-degree relative with autopsy-proven NDLVC) is highly suggestive of NDLVC and warrants close follow-up.

In the absence of conclusive genetic information in the family, NDLVC should be considered familial if one or more first- or second-degree relatives have NDLVC, or when SCD has occurred in a first-degree relative at any age with an established diagnosis of NDLVC. Familial disease should also be suspected if a first-degree relative has sudden death at <50 years of age and autopsy findings suggestive of the NDLVC phenotype.

7.3.1.3. Diagnostic work-up

The key elements of the diagnostic work-up for all patients with NDLVC are described in *Section 6* and include clinical history, laboratory tests, Holter monitoring and cardiac imaging, and genetic testing.

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Echocardiography and CMR are both central to the diagnosis. Additional laboratory tests, exercise testing, EMB, and cardiac catheterization may also be considered (see *Section 6*).

7.3.1.4. Electrocardiographic features

Recommendations on resting and ambulatory ECG testing are described in *Section 6.5* and are of particular importance in patients with NDLVC, as specific features can indicate the underlying genetic cause. Prolonged PR interval or AV block is frequent in neuromuscular causes of NDLVC and in sarcoidosis. Laminopathies are characterized by prolonged PR interval, AF, and ventricular ectopics (VEs), and frequently show low voltage in pre-cordial leads.[887] Depolarization abnormalities such as low QRS voltage are also a common finding in NDLVC caused by *DSP* and *PLN* variants.[542] Ambulatory ECG monitoring is useful in NDLVC patients to reveal supraventricular and ventricular arrhythmias or bradycardias due to AV conduction block and is recommended at least yearly, or when there is a change in clinical status. In some patients with NDLVC at high risk of developing conduction disease and/or arrhythmias (including laminopathies, neuromuscular disease, *PLN*, and *FLNC*-truncating variants), Holter monitoring may be considered more frequently.

Recommendation Table 25 — Recommendation for resting and ambulatory electrocardiogram monitoring in patients with non-dilated left ventricular cardiomyopathy

Recommendation	Class[a]	Level[b]
Ambulatory ECG monitoring is recommended in patients with NDLVC annually or when there is a change in clinical status, to aid in management and risk stratification.	I	C

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ECG, electrocardiogram; NDLVC, non-dilated left ventricular cardiomyopathy.
[a]Class of recommendation.
[b]Level of evidence.

7.3.1.5. Echocardiography

Comprehensive TTE is recommended for all patients with NDLVC as it provides all relevant information on the global and regional LV anatomy, function, and haemodynamics; valvular heart disease; right heart function; pulmonary pressure; atrial geometry; and other features.[71,73] Advanced echocardiographic techniques (including deformation imaging using tissue Doppler and speckle tracking) can allow the early detection of subclinical myocardial dysfunction in specific situations (e.g. genetic NDLVC carriers).[71,74]

7.3.1.6. Cardiac magnetic resonance

Cardiac magnetic resonance with LGE is the foremost imaging modality in NDLVC as it provides confirmation of the presence of non-ischaemic myocardial fibrosis that is essential for the diagnosis in most cases. Cardiac magnetic resonance can also be useful to detect the presence of myocardial oedema, which may suggest an inflammatory or myocarditic aetiology, and to describe the extent and pattern of fibrosis distribution. This can provide clues to the underlying aetiology (e.g. subepicardial distribution in post-myocarditis forms, patchy in sarcoidosis, extensive inferolateral in dystrophinopathies, septal mid-wall in *LMNA* carriers, and ring-like in *DSP* and *FLNC* variant carriers),[71] and may provide additional prognostic value both for arrhythmia and heart failure severity.[137,848]

7.3.1.7. Nuclear medicine

The role of radionuclide imaging in NDLVC is limited. Measurement of 18F-FDG uptake using PET, with focal or focal-on-diffuse FDG uptake patterns, especially if concomitant abnormal 18F-FDG-PET uptake in extracardiac tissues, can be useful in suspected cardiac sarcoidosis.[849] Isolated cardiac uptake has also been described in patients with NDLVC caused by *DSP* variants.[888]

7.3.1.8. Endomyocardial biopsy

Endomyocardial biopsy (EMB) with immunohistochemical quantification of inflammatory cells remains the gold standard investigation for the identification of cardiac inflammation. It may confirm the diagnosis of autoimmune disease in patients with unexplained heart failure and suspected giant cell myocarditis, eosinophilic myocarditis, vasculitis, and sarcoidosis. In experienced centres, electroanatomical voltage mapping-guided EMB may improve the yield of diagnosis of NDLVC.[889] The risks and benefits of EMB should be evaluated and this procedure should be reserved for specific situations where its results may affect diagnosis or treatment (see *Section 6.7.5*).

7.3.2. Genetic testing

The genes most commonly implicated in NDLVC are *DSP*, *FLNC* (truncating variants), *DES*, *LMNA*, or *PLN*, but there is substantial overlap with the genetic background of both DCM and ARVC (*Table 10*). Desmoplakin (*DSP*) variants, in particular, cause a unique form of cardiomyopathy with a high prevalence of LV fibrosis and myocardial inflammatory episodes.[864]

The identification of a P/LP gene variant in a patient with NDLVC allows better prediction of the disease outcome and progression, may contribute to the indications for device implantation, informs genetic counselling, and allows familial screening for relatives (see *Section 6.8.3*). Therefore, genetic testing is recommended in all patients with NDLVC.

Recommendations for clinical screening, genetic counselling, and testing are described in *Sections 6.8.3* and *6.11*. Evaluation of conduction disease, and atrial and ventricular arrhythmia, is of particular importance in patients with NDLVC, as these may often be early phenotypic features. There are very few data on the natural history of phenotype-negative variant carriers or on the clinical yield of familial cascade screening in NDLVC, but cross-sectional studies suggest age-related increases in penetrance.[9] Precautionary long-term evaluation of first-degree relatives is therefore recommended.

7.3.3. Assessment of symptoms

Most patients with NDLVC are asymptomatic, but some develop symptoms related to arrhythmia or conduction disease (e.g. syncope, palpitation) or diastolic heart failure (e.g. dyspnoea). Sustained ventricular arrhythmia, cardiac arrest, or SCD can be the initial presentation in a proportion of patients. Assessment of symptoms in patients with cardiomyopathies is described in *Section 6.10.1*.

7.3.4. Management

The clinical management of heart failure and other manifestations of NDLVC (atrial tachyarrhythmia, conduction disease) has been described in the *2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure*, the *2020 ESC Guidelines for the diagnosis and management of atrial fibrillation*, the *2021 ESC Guidelines on cardiac*

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pacing and cardiac resynchronization therapy, and the *2022 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death*,[69,299,336,724] and are discussed in *Section 6.10.2*.

7.3.5. Sudden cardiac death prevention in non-dilated left ventricular cardiomyopathy

The prediction and prevention of SCD is the cornerstone of the clinical care of patients with NDLVC.

7.3.5.1. Secondary prevention of sudden cardiac death

As in other cardiomyopathy subtypes, ICD implantation is recommended in survivors of cardiac arrest and in patients who have experienced sustained ventricular arrhythmias with haemodynamic compromise;[531] the decision to implant should consider the patient's view and their QoL, as well as the absence of other diseases likely to cause death within 1 year.

7.3.5.2. Primary prevention of sudden cardiac death

There are no available RCTs examining the usefulness of ICDs to prevent SCD in patients with mild or moderate LV dysfunction. Recommendations for ICD implantation in DCM individuals with LVEF <35% are discussed in *Section 6.10.2* and also apply to patients with NDLVC and LVEF <35%. Most patients with NDLVC, however, have either normal or mildly impaired LV systolic function. Much of the data on the natural history and risk prediction in NDLVC are derived from cohorts that include either patients with DCM or with ARVC (see *Sections 7.2* and *7.4*), and data on patients with NDLVC are therefore necessarily very limited. However, the available data suggest that genotype is a major determinant of SCD risk, with patients harbouring variants in *PLN*, *TMEM43*, *DES*, *DSP*, *LMNA*, *FLNC* (truncating variants), and *RBM20* having a substantially higher rate of major arrhythmic events than other causes regardless of LVEF.[440,542,864–869] For some high-risk genotypes (e.g. *LMNA* [https://lmna-risk-vta.fr][541]), gene-specific (or, in the case of the *PLN* p.Arg14del variant, variant-specific [https://plnriskcalculator.shinyapps.io/final_shiny])[542] risk-prediction scores have been developed that consider genotype characteristics and additional phenotypic features. Where such scores are available, they should be used to guide primary prevention ICD implantation (*Figure 12*). As discussed in *Sections 7.1.5* and *7.2.5*, the Task Force acknowledges the challenges associated with defining universal thresholds for acceptable risk across different cardiomyopathy phenotypes, but is of the opinion that a similar approach to that taken in risk stratification for HCM is reasonable, although the *2022 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death* have suggested a higher threshold of 10% risk at 5 years to guide primary prevention ICD implantation in patients with NDLVC and *LMNA* variants.[3] This Task Force recommends more overt shared decision-making based on real-world data as well as individual preferences, beliefs, circumstances, and values. Gaps in evidence are acknowledged and should be shared with patients, and competing risks related to the disease (heart failure, stroke) and to age and comorbidity, as well as device-related complications, should be discussed.

There are very few data to guide risk stratification in patients with NDLVC without a known causative gene variant, but on the basis of the existing literature, the Task Force suggests that it may be reasonable to consider primary prevention ICD implantation in patients with NSVT, a family history of SCD, or significant LGE. Additional risk factors, such as the burden of VE, may also help guide ICD implantation, but there are no data currently to support a specific threshold for VE burden, and this will depend on the underlying genotype and other clinical factors.[542,867,872] In patients with unexplained syncope, PES may provide additional information on the underlying cause.[875] There are no definitive data supporting the regular use of PES for primary prevention risk stratification in patients with NDLVC, but may be beneficial in patients with NDLVC and myotonic dystrophy with an independent indication to EP study to assess conduction disturbances,[876] although the clinical value of this approach has not been consistently demonstrated.[877] Given the overlap with DCM and available data, and in keeping with the *2022 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death*, the Task Force agreed that the recommendations for primary prevention ICD implantation in NDLVC should be the same as those for DCM (*Figure 17*), but the level of evidence is necessarily lower.

Recommendation Table 26 — Recommendations for an implantable cardioverter defibrillator in patients with non-dilated left ventricular cardiomyopathy

Recommendations	Class[a]	Level[b]
Secondary prevention		
An ICD is recommended to reduce the risk of sudden death and all-cause mortality in patients with NDLVC who have survived a cardiac arrest or have recovered from a ventricular arrhythmia causing haemodynamic instability.	I	C
Primary prevention		
An ICD should be considered to reduce the risk of sudden death and all-cause mortality in patients with NDLVC, heart failure symptoms, and LVEF ≤35% despite >3 months of OMT.[861,885]	IIa	A
The patient's genotype should be considered in the estimation of SCD risk in NDLVC.	IIa	C
An ICD should be considered in patients with NDLVC with a genotype associated with high SCD risk and LVEF >35% in the presence of additional risk factors (see *Table 21*).[185,186,438,541,542,865–869,878–883]	IIa	C
An ICD may be considered in selected patients with NDLVC with a genotype associated with high SCD risk and LVEF >35% without additional risk factors (see *Table 21*).	IIb	C
An ICD may be considered in patients with NDLVC without a genotype associated with high SCD risk and LVEF >35% in the presence of additional risk factors.[c]	IIb	C

ICD, implantable cardioverter defibrillator; LVEF, left ventricular ejection fraction; NDLVC, non-dilated left ventricular cardiomyopathy; OMT, optimal medical therapy; SCD, sudden cardiac death.
[a]Class of recommendation.
[b]Level of evidence.
[c]Additional risk factors include syncope, LGE presence on CMR.

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7.4. Arrhythmogenic right ventricular cardiomyopathy

7.4.1. Diagnosis

7.4.1.1. Index case

Arrhythmogenic right ventricular cardiomyopathy (ARVC) is characterized structurally by a progressive myocardial atrophy with fibro-fatty replacement of the RV myocardium.[890] Lesions can also be present in the LV myocardium; predominant LV disease can coexist in the same family. Arrhythmogenic right ventricular cardiomyopathy usually manifests in the second to fourth decade of life.[891] Men are affected more frequently than women and an age-related penetrance has been demonstrated, with high clinical and genetic variability.

An ARVC diagnosis should be suspected in adolescents or young adults with palpitations, syncope, or aborted sudden death. Frequent VEs or VT of LBBB morphology are among the most common clinical presentation. The presence of right pre-cordial TWI (V1–V3) in routine ECG testing should also be suspected for ARVC.[10,892] Less common ECG changes include low QRS voltages in the peripheral leads and terminal activation delay in the right pre-cordial leads.[893] Right ventricular dilatation on 2D echocardiography is also a frequent reason for patient referral. Less common presentations are RV or biventricular heart failure that can mimic DCM or NDLVC.[894] Patients with multiple variants are thought to develop a more severe phenotype, and patients with a *DSP* or *DSG2* variant are more prone to develop heart failure.[895,896]

In children and young adults, syncope, palpitations, and ventricular arrhythmias are also the usual presenting symptoms.[897] However, chest pain, dynamic ST-T wave changes on basal 12-lead ECG, and myocardial enzymes release in the setting of normal coronary arteries are often reported, requiring differential diagnosis with myocarditis or acute myocardial infarction.[898]

7.4.1.2. Relatives

Clinical testing in relatives often reveals non-diagnostic abnormalities. In this context, the presence of RV systolic global or regional dysfunction, or additional electrocardiographic abnormalities (e.g. repolarization abnormalities, prolonged terminal activation duration, low QRS voltages, frequent ventricular extrasystoles [>500 per 24 h], or NSVT) in a first-degree relative of an individual with ARVC (or a first-degree relative with autopsy-proven ARVC) is highly suggestive of ARVC and warrants close follow-up.

7.4.1.3. Diagnostic work-up

The key elements of the diagnostic work-up for all patients with ARVC are defined by the diagnostic criteria used for the identification of affected individuals. The revised Task Force criteria for the diagnosis of ARVC published by Marcus *et al.* in 2010 have been used for the diagnosis of ARVC for more than a decade.[10] More recently, the Padua criteria have offered an updated iteration to include LV involvement but are yet to be externally validated.[5] Key elements of the diagnostic work-up include ECG, Holter monitoring, cardiac imaging, genetic testing, and, in specific circumstances, EMB.[4,10,892] Additional laboratory tests, exercise testing, and cardiac catheterization should also be considered, as detailed in *Section 6*.

7.4.1.4. Electrocardiography and Holter monitoring

Abnormalities of the repolarization and depolarization as well as arrhythmias are key to the diagnosis of ARVC.[5] The diagnostic utility of late potentials on signal-averaged electrocardiogram (SAECG) has been challenged in patients with ARVC for showing poor sensitivity and specificity.[5,899] It has been noted that epsilon waves are frequently overdiagnosed and that there is poor agreement even between experts regarding their presence.[900] Furthermore, it has been demonstrated that they occur in the presence of severe structural disease and thus add little to the diagnosis.[900,901] Therefore, epsilon waves and SAECG should be utilized for diagnostic purposes with caution.

Recommendation Table 27 — Recommendation for resting and ambulatory electrocardiogram monitoring in patients with arrhythmogenic right ventricular cardiomyopathy

Recommendation	Class[a]	Level[b]
Annual ambulatory ECG monitoring is recommended in patients with ARVC to aid in diagnosis, management, and risk stratification.[902]	I	C

ARVC, arrhythmogenic right ventricular cardiomyopathy; ECG, electrocardiogram.
[a]Class of recommendation.
[b]Level of evidence.

7.4.1.5. Echocardiography and cardiac magnetic resonance

A comprehensive cardiac imaging assessment is recommended for all ARVC patients.[71,73] Structural and functional alterations assessed by echocardiography and CMR are key to ARVC diagnosis.[10] The key feature is the presence of wall motional abnormalities such as RV akinesia, dyskinesia, or bulging, and the determinant of the diagnostic performance is the level of RV dilatation or dysfunction (major and minor criteria). Cardiac magnetic resonance should be considered the first-line test for assessment of the RV functional structural abnormalities criterion as it has demonstrated superior sensitivity.[10] Contrast-enhanced CMR is the only tool allowing the detection of LV involvement which remains otherwise underestimated by applying the 2010 Task Force Criteria. Tissue characterization by CMR or indirectly by electroanatomical voltage mapping may show signs of fibro-fatty replacement that can be present in either ventricle.[889,903,904]

7.4.1.6. Endomyocardial biopsy

Differential diagnosis in patients with suspected ARVC includes inflammatory processes affecting the right ventricle such as myocarditis and sarcoidosis. In some instances, especially when dealing with probands with a sporadic form, EMB may be helpful to rule out myocarditis and sarcoidosis.[72,892,905] Endomyocardial biopsy can also be useful in selected patients in whom non-invasive assessment is inconclusive.[4,72] Electroanatomic voltage mapping-guided EMB may be considered in selected cases, particularly in case of negative CMR.[906]

7.4.1.7. Nuclear medicine

Measurement of 18F-FDG uptake using PET, with focal or focal-on-diffuse FDG uptake patterns, can be useful in suspected cardiac sarcoidosis.[849] However, it has been demonstrated that patients with ARVC can also show myocardial 18F-FDG-PET uptake.[888,907] Therefore, there is a limited role for radionuclide imaging in ARVC unless there is concomitant abnormal 18F-FDG-PET uptake in extracardiac tissues, or other clinical features suggestive of cardiac sarcoidosis.[904,908]

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7.4.1.8. Arrhythmogenic right ventricular cardiomyopathy phenocopies

In suspicion of ARVC, a systematic approach to investigation of phenocopies should be undertaken. Differential diagnosis in patients with suspected ARVC includes myocarditis, sarcoidosis, RV infarction, DCM, Chagas disease, pulmonary hypertension, and CHD with volume overload (such as Ebstein anomaly, atrial septal defect, and partial anomalous venous return, left-to-right shunt, and pericardial agenesis).[909,910] Disease phenocopies also include non-structural diseases. In fact, one of the main diagnostic dilemmas is to distinguish ARVC from idiopathic RV outflow tract VT, since the latter is usually benign.[4] The idiopathic nature of VT is supported by the absence of family history, a normal basal 12-lead ECG, a normal ventricular structure by cardiac imaging and electroanatomic mapping, a single VT morphology, and the non-inducibility at programmed ventricular stimulation.

In highly trained competitive athletes, differential diagnosis with physiological adaptation to training needs to be considered.[911] Right ventricular enlargement, ECG abnormalities, and arrhythmias reflect the increased haemodynamic load during exercise. While global RV systolic dysfunction and/or RWMAs, such as bulgings or aneurysms, are more in keeping with ARVC, the absence of overt structural changes of the right ventricle, frequent VEs, or inverted T waves in pre-cordial leads all support a benign nature (so-called athlete's heart).[72,912,913]

7.4.2. Genetic testing and family screening

The genes underlying ARVC mainly encode proteins of the cardiac desmosome: plakophilin-2 (*PKP2*), desmoplakin (*DSP*), desmoglein-2 (*DSG2*), desmocollin-2 (*DSC2*), and plakoglobin (*JUP*). In addition to desmosomal genes, P/LP variants have also been described in other genes, including *DES*,[914] *TMEM43*,[915] and *PLN*.[190,882] Pathogenic or likely pathogenic variants can be identified in up to 60% of patients with a diagnosis of ARVC.[230] Given the diagnostic importance of genetic testing in ARVC, it is important that genetic variants are frequently re-appraised in terms of their pathogenicity.[916] The pattern of inheritance in the majority of ARVC families is autosomal dominant. The penetrance of the disease in genetic carriers is age, gender, and physical activity dependent.[892,917]

Recommendations for clinical screening, genetic counselling, and testing are described in *Sections 6.8.3* and *6.11*. Cardiac evaluation should be adapted to the particular risk of complications in the family. Evaluation every 1–2 years including ECG, ECHO, and Holter/ECG monitoring is generally recommended for relatives at risk of developing the disease. Cardiac magnetic resonance should be considered at the baseline evaluation.

7.4.3. Assessment of symptoms

Patients with ARVC commonly experience palpitations and can develop symptoms of heart failure, although this may occur many years after the appearance of the initial abnormalities. Assessment of symptoms in patients with cardiomyopathies is described in *Section 6.10.1*.

7.4.4. Management

The aim of the clinical management of ARVC relies on the improvement of symptoms, the reduction of the pace of disease progression, and the prevention of complications. Recommendations for the pharmacological management of atrial arrhythmias and heart failure symptoms in patients with cardiomyopathies are described in *Sections 6.10.2* and *6.10.3*.

7.4.4.1. Antiarrhythmic therapy

Beta-blockers constitute the first option to reduce arrhythmic burden via a reduction in adrenergic tone, particularly on exercise. Titration to the maximal tolerated dose has been associated with an improvement in survival from major ventricular arrhythmias in retrospective observational studies.[918]

Amiodarone is often used when other beta-blockers fail to control arrhythmias.[917,919,920] It should, however, be used with caution for the long-term management of ventricular arrhythmias, especially in young patients. Sotalol has been used for many years, but evidence regarding its efficacy remain limited and conflicting.[921,922] Flecainide should be considered when single agent treatment has failed to control arrhythmia-related symptoms in patients with ARVC or when autonomic side effects limit the use of beta-blockers.[923,924] Experience with other antiarrhythmics (dofetilide, ranolazine) is limited to very small case series.[919,923]

A proportion of patients require invasive arrhythmic procedures and/or ICD implantation. Complex endocardial and/or epicardial approach guided by three-dimensional (3D) electroanatomical mapping can be recommended but with a high recurrence rate (30–50% in experienced centres).[919,925–927] Sympathetic denervation has also been used.[928] Such procedures do not confer adequate protection against SCD, but may be very useful in reducing the VT burden and the risk of electrical storm.[917] Discontinuation of intense physical exercise has shown a potential to slow the pace of disease progression and reduce the ventricular arrhythmia burden.[917,919]

Recommendation Table 28 — Recommendations for the antiarrhythmic management of patients with arrhythmogenic right ventricular cardiomyopathy

Recommendations	Class[a]	Level[b]
Beta-blocker therapy is recommended in ARVC patients with VE, NSVT, and VT.[920–922]	I	C
Amiodarone should be considered when regular beta-blocker therapy fails to control arrhythmia-related symptoms in patients with ARVC.[921,922]	IIa	C
Flecainide in addition to beta-blockers should be considered when single agent treatment has failed to control arrhythmia-related symptoms in patients with ARVC.[923,924]	IIa	C
Catheter ablation with availability for epicardial approach guided by 3D electroanatomical mapping of VT should be considered in ARVC patients with incessant VT or frequent appropriate ICD interventions for VT despite pharmacological therapy with beta-blockers.[925,929–934]	IIa	C

3D, three-dimensional; ARVC, arrhythmogenic right ventricular cardiomyopathy; ICD, implantable cardioverter defibrillator; NSVT, non-sustained ventricular tachycardia; VE, ventricular ectopic beats; VT, ventricular tachycardia.
[a]Class of recommendation.
[b]Level of evidence.

7.4.5. Sudden cardiac death prevention in arrhythmogenic right ventricular cardiomyopathy

Arrhythmogenic right ventricular cardiomyopathy is characterized by its high propensity for ventricular arrhythmias and SCD.[919] Although estimated to be a rare disease, it has been consistently reported as one of

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the most common causes of SCD in registries around the world.[935–937] Sudden cardiac death seems to be more prevalent in young athletic individuals affected by the disease.[935,938]

7.4.5.1. Secondary prevention of sudden cardiac death

Implantable cardioverter defibrillators reduce mortality in survivors of cardiac arrest and in patients who have experienced sustained symptomatic ventricular arrhythmias.[531] An ICD is recommended in such patients when the intent is to increase survival; the decision to implant should consider the patient's view and their QoL, as well as the absence of other diseases likely to cause death within the following year.

7.4.5.2. Primary prevention of sudden cardiac death

Most of the current evidence on the outcomes of patients with ARVC and their predictors is limited to observational retrospective cohort studies that are typically of small size.[939] Thus, the number of clinical predictors that can be studied using multivariate models is very limited, and most studies cannot be compared with one another. A systematic review and meta-analysis ($n = 18$ studies) has shown that the average risk of ventricular arrhythmia ranges from 3.7 to 10.6% per year and that male sex, RV dysfunction, and prior non-sustained or sustained VT/VF consistently predict ventricular arrhythmias in populations with ARVC.[939]

The first comprehensive effort to offer an approach to risk stratification in the context of decision-making for ICD implantation was made in the 2015 International Task Force consensus statement (ITFC) on the treatment of ARVC/dysplasia, where recommendations were made according to the presence of risk factors that would characterize the risk level of each patient.[919] A follow-up study ($n = 365$) offered a modification on the International Task Force approach that resulted in better discrimination.[940] The *2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death*[941] and the *2019 HRS expert consensus statement on evaluation, risk stratification, and management of arrhythmogenic cardiomyopathy*[4] have also offered alternative approaches to this issue. A risk-prediction model was developed from a multicentre collaboration ($n = 528$); it utilizes sex, age, recent syncope, NSVT, VE count, number of leads with TWI, and right ventricular ejection fraction (RVEF) as predictors to provide an individualized estimate of sustained ventricular arrhythmias in patients with ARVC (arvcrisk.com).[539]

A study ($n = 617$) comparing the previous approaches to risk stratify patients has revealed that the modified ITFC approach provides the highest net benefit, up to an estimated 5-year risk of 25%, whereas AHA and HRS perform best in patients with an estimated 5-year risk >25%.[538] In the same study, an estimated 5-year risk of 12.5% seems to be the optimal threshold, beyond which the risk-prediction model has the best performance. An external comparison ($n = 140$) of the different ARVC risk levels showed that the highest net benefit was seen with a 10% cut-off, using the 2019 ARVC risk calculator.[536] In the same study, the 10% cut-off was superior to the HRS and ITFC approaches.[536] Another external validation study ($n = 128$) of the 2019 ARVC risk model showed that although discriminative ability is excellent (c-index 0.84), the model seems to significantly overestimate the risk of patients below the 50% 5-year risk threshold.[537] Recently, a correction to the 2019 ARVC risk calculator was issued.[539] Two large external validation studies of the updated 2019 ARVC risk calculator have been published, suggesting a good discriminative performance, but the latter study revealed overestimation of risk.[524,526] This raises concerns regarding the accuracy of the model in offering an individualized prediction that can help inform patients during decision-making; however, it can remain informative due to its excellent discriminative performance. Furthermore, one study suggested that the updated 2019 ARVC risk calculator performs best in *PKP2* patients, but its performance is more limited in gene-negative individuals.[524]

Therefore, a combination of these approaches is recommended to individualize risk quantification that can aid clinicians in balancing the risks and benefits of ICD implantation. The final decision should be made together with the patient, considering other competing risks and the patient's risk tolerance. As discussed in *Section 7.1.5*, the Task Force acknowledges the challenges associated with defining universal thresholds for acceptable risk across different cardiomyopathy phenotypes, but is of the opinion that a similar approach to that taken in risk stratification for HCM, DCM, and NDLVC is reasonable. In this context, the Task Force recommends shared decision-making based on real-world data as well as individual preferences, beliefs, circumstances, and values. Gaps in evidence should be shared with patients, and competing risks related to the disease (heart failure, stroke) and to age and comorbidity, as well as device-related complications, should be discussed. The suggested approach is summarized in *Figure 18*.

Patients with ARVC are known to suffer from sustained VTs that can be well tolerated without leading to SCD. Using appropriate ICD interventions as surrogate for SCD outcome has been shown to overestimate SCD.[942] Considering that, in most centres, a high ratio of ARVC patients will be implanted with an ICD, it is conceivable why this may hamper risk stratification for SCD in patients with ARVC. Efforts to address this have been made within several studies,[522,523,943–947] where

Recommendation Table 29 — Recommendations for sudden cardiac death prevention in patients with arrhythmogenic right ventricular cardiomyopathy

Recommendations	Class[a]	Level[b]
Secondary prevention		
An ICD is recommended to reduce the risk of sudden death and all-cause mortality in patients with ARVC who have survived a cardiac arrest or have recovered from a ventricular arrhythmia causing haemodynamic instability.[939,943,944,948,949]	I	A
An ICD should be considered in ARVC patients who have suffered a haemodynamically tolerated VT.[522,939,943–945,948–950]	IIa	B
Primary prevention		
High-risk features[c] should be considered to aid individualized decision-making for ICD implantation in patients with ARVC.[538,939]	IIa	B
The updated 2019 ARVC risk calculator should be considered to aid individualized decision-making for ICD implantation in patients with ARVC.[d,524,526,536–539]	IIa	B

ARVC, arrhythmogenic right ventricular cardiomyopathy; ICD, implantable cardioverter defibrillator; LVEF, left ventricular ejection fraction; NSVT, non-sustained ventricular tachycardia; PES, programmed electrical stimulation; RVEF, right ventricular ejection fraction; SMVT, sustained monomorphic ventricular tachycardia; VT, ventricular tachycardia.

[a]Class of recommendation.

[b]Level of evidence.

[c]High-risk features: arrhythmic syncope, NSVT, RVEF <40%, LVEF <45%, SMVT at PES.

[d]See text for discussion of gene-specific differences in the performance of the updated 2019 ARVC risk calculator.

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the outcome of interest is fast VT (>250 b.p.m.) rather than any sustained VT. The largest of these studies ($n = 864$) has led to the development of a separate score for the prediction of unstable VT/VF.[945] Due to the lack of any external validation studies, there is insufficient information to support the applicability of this risk score outside of its development cohorts. Furthermore, a specific rate cut-off is also not well evidence-based and its performance to predict SCD remains unclear. Although it is likely that slower sustained VTs per se are not life-threatening, it remains unknown how frequently they would degenerate to faster VTs or VF. It is therefore reasonable to suggest that all patients at risk of any sustained ventricular arrhythmia should be offered primary prevention ICDs.

The role of PES in risk stratification of ARVC patients is not well defined, particularly in those who are asymptomatic.[523,939] However, current practice suggests that inducibility of SMVT at PES might add value in patients with symptoms consistent with sustained ventricular arrhythmia and this is further supported in this guideline[3]

7.5. Restrictive cardiomyopathy

7.5.1. Diagnosis

Patients with overt RCM manifest signs and symptoms typical of HFpEF.[306] The systematic approach to diagnosis should include clinical examination, ECG, echocardiography, and CMR.[951] Physical examination may show a prominent jugular venous pulse. In the advanced phases, the pulse volume is low, the stroke volume declines, and the heart rate may increase. Hepatomegaly, ascites, and peripheral oedema are common in decompensated patients. Echocardiography is the gold standard diagnostic tool; criteria for diagnosing and grading diastolic dysfunction have been previously described.[951,952] Importantly, the degree of diastolic dysfunction in patients with RCM is often only truly restrictive in advanced stages and most patients show milder grades of diastolic impairment at diagnosis.[951] Cardiac catheterization should be performed in cases where the diagnosis is in doubt and to aid in the assessment for and timing of cardiac transplantation.[953] Cardiac MRI distinguishes RCM from constrictive pericarditis, provides information on the presence and extent of myocardial fibrosis, and contributes to distinguishing metabolic from inflammatory diseases.[951,954] Endomyocardial biopsy is a precision diagnostic tool in restrictive cardio-desminopathies;[955] iron myocardial overload, both intramyocyte in haemochromatosis[956] and mitochondrial in Friedreich ataxia cardiomyopathy;[957] cystinosis;[958] generalized arterial calcification of infancy;[955,959] and lysosomal storage diseases (LSDs).[960,961] Deep phenotyping in probands should go beyond cardiac traits and explore extracardiac manifestations in syndromic diseases and in RCM associated with neuromuscular disorders (see *Section 6*).[962]

7.5.2. Genetic testing

When inherited, RCM most commonly presents as an autosomal dominant disorder and, less commonly, autosomal recessive or sporadic. Genes associated with RCM encode sarcomeric structural and regulatory proteins and cytoskeletal intermediate filaments (*Table 10*). Although all major sarcomeric genes may cause RCM,[963] the most common disease gene is *TNNI3*, which encodes the thin filament troponin I.[964] Other less commonly involved genes include *TNNT2*, *ACTC1*, *MYH7*, *MYBPC3*, *TTN*, *TPM1*, *MYPN*, *MYL3*, and *MYL2*. Restrictive cardiomyopathy can be associated with intramyocyte accumulation of unfolded defective proteins, a feature that is increasingly demonstrated in carriers of defects in *DES*, *FLNC*, and *BAG3*. These diseases have significant implications for prognosis and timely decision-making, both in children and adults. Restrictive cardiomyopathy may also occur in individuals with a family history of HCM[289] or DCM.[965] The observation of different cardiomyopathy phenotypes within families suggests a variable response to the variant, and

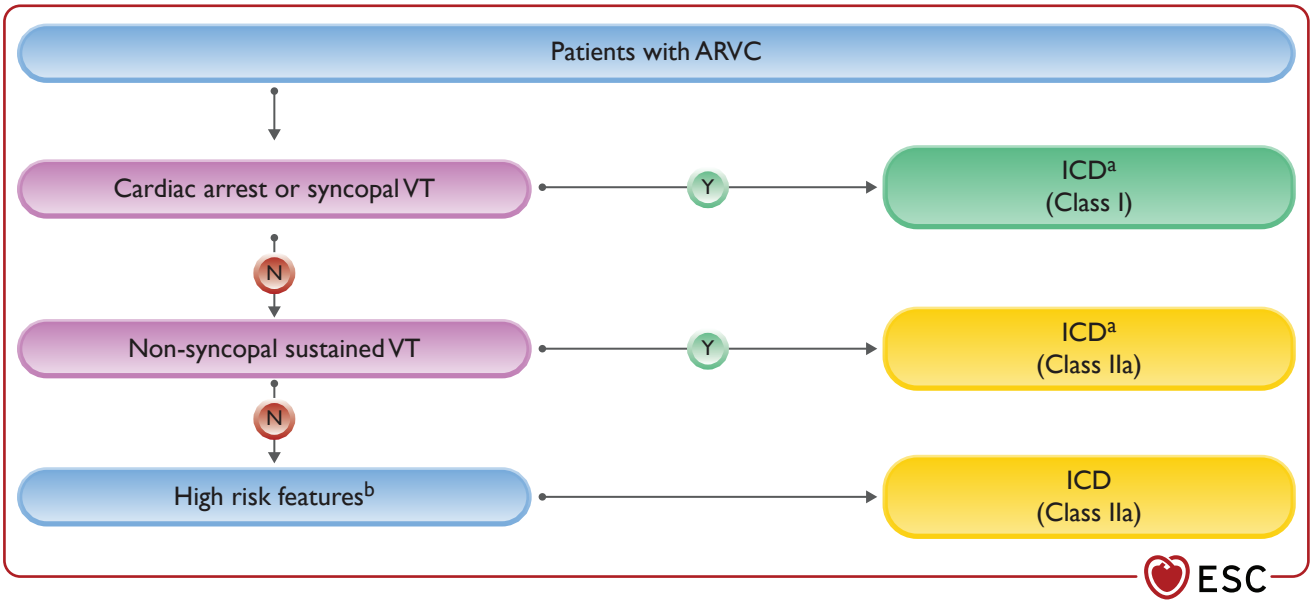


Figure 18 Algorithm to approach implantable cardioverter defibrillator decision-making in patients with arrhythmogenic right ventricular cardiomyopathy. ARVC, arrhythmogenic right ventricular cardiomyopathy; ICD, implantable cardioverter defibrillator; LVEF, left ventricular ejection fraction; NSVT, non-sustained ventricular tachycardia; PES, programmed electrical stimulation; RVEF, right ventricular ejection fraction; SMVT, sustained monomorphic ventricular tachycardia; VT, ventricular tachycardia. [a]Clinicians should aim to control ventricular arrhythmia with pharmacological or invasive antiarrhythmic therapies in addition to offering an ICD. [b]High-risk features are defined as either cardiac syncope, NSVT, RVEF <40%, LVEF <45%, SMVT at PES or as per the updated 2019 ARVC risk calculator.[539]

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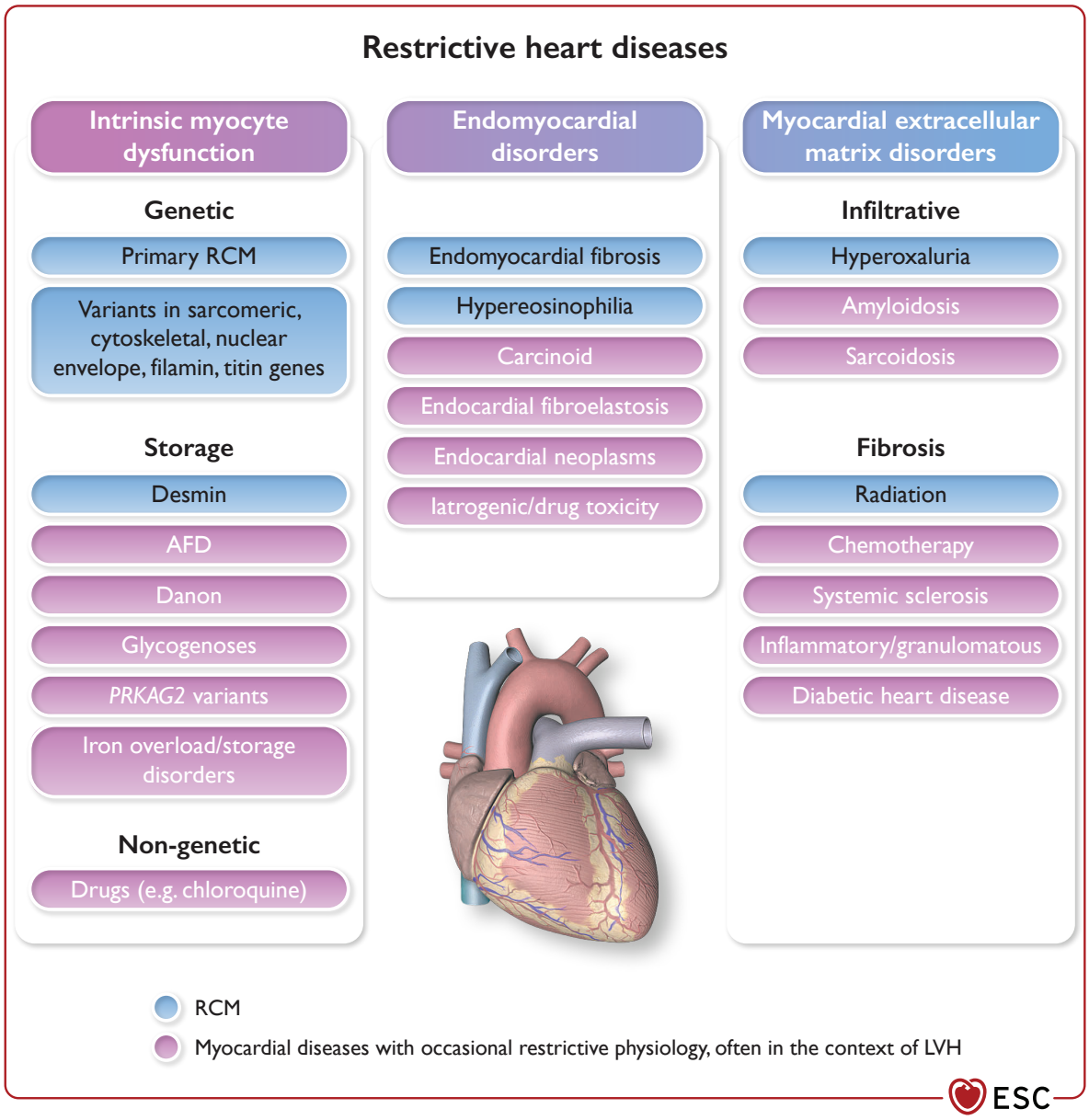


Figure 19 Spectrum of restrictive heart diseases. AFD, Anderson–Fabry disease; LVH, left ventricular hypertrophy; PRKAG2, Protein kinase AMP-activated non-catalytic subunit gamma 2; RCM, restrictive cardiomyopathy. For a more detailed spectrum of restrictive heart disease, please refer to Supplementary data online, *Table S4*.

implicates factors beyond the specific variant in the determination of ultimate clinical manifestation of disease.[966] Hereditary infiltrative diseases can also cause RCM, the most common of which is amyloidosis caused by pathogenic variants in the *TTR* gene, although this is usually in the presence of LVH (see *Section 7.7*).

7.5.3. Assessment of symptoms

Patients with RCM often develop symptoms of heart failure, although this can occur some years after the appearance of the initial abnormalities. Assessment of symptoms in patients with cardiomyopathies is described in *Section 6.10.1*.

7.5.4. Management

The administration of heart failure medications and device implantation, including ventricular assist device as a bridge-to-candidacy is guided by symptoms and heart failure phenotype and severity,[967] and is described in *Section 6.10.2*. Precision diagnosis (phenotype and cause) is key to timely planning of heart transplantation as it guarantees the exclusion of all genetic and acquired phenocopies that may be amenable to alternative treatment. Prevention of heart transplantation in all RCM patients with alternative treatments is a major goal for all adult and paediatric RCM.

Precise diagnosis is also essential for genetic phenocopies with available target treatments: ERT for Anderson–Fabry disease or glycogenosis such as Pompe disease; therapeutic phlebotomy for haemochromatosis; immunosuppressive therapeutics for sarcoidosis; new biological drugs for systemic diseases (e.g. autoimmune diseases with cardiac involvement that can reverse or stabilize by treating the disease itself); and removal of the toxic causes (see *Figure 19* and Supplementary data online, *Table S4*). Precision diagnosis today is essential due to the increasing availability of disease-specific treatments and diagnostic tools to exclude geno/phenocopies.

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Restrictive cardiomyopathy is associated with the worst prognosis of all the cardiomyopathy phenotypes. Survival data are limited to small windows of observation. The prognosis of RCM largely depends on the restrictive physiology, regardless of the underlying cause.[968–971] More than 50% of children with RCM are at risk of death (including SCD) or transplantation shortly after diagnosis; clinical features putatively associated with increased risk of death or transplantation include: heart failure symptoms; reduced LV systolic function; increased left atrial size; syncope; ischaemia; and impaired LV diastolic function on echocardiography.[286,969,972,973] Up to 75% of surviving patients demonstrate heart failure, and the outcome is either death or heart transplantation within a few years of diagnosis.[968,969] Elevated pulmonary vascular resistance (PVR) is present in up to 40% of children with RCM, and can rise quickly even in the absence of other clinical changes, which has an impact on suitability for and timing of cardiac transplantation.[953] Cardiac catheterization with an assessment of PVR is therefore recommended in all children at diagnosis and every 6 to 12 months.[953] In adult patients with genetic RCM, the main cause of death is heart failure (more than 40%), with a 5-year survival rate of ~50% in cohorts that include patients with HCM and restrictive physiology.[616]

Recommendation Table 30 — Recommendations for the management of patients with restrictive cardiomyopathy

Recommendations	Class[a]	Level[b]
It is recommended that multimodality imaging be used to differentiate RCM from HCM or DCM with restrictive physiology.	I	C
It is recommended that baseline cardiac and non-cardiac investigations are performed to assess involvement of the neuromuscular system or other syndromic disorders.	I	C
Cardiac catheterization is recommended in all children with RCM to measure pulmonary artery pressures and PVR at diagnosis and at 6–12 monthly intervals to assess change in PVR.[953]	I	B
ICD implantation is recommended to reduce the risk of sudden death and all-cause mortality in patients with RCM who have survived a cardiac arrest or have recovered from a ventricular arrhythmia causing haemodynamic instability.	I	C
Endomyocardial biopsy should be considered in patients with RCM to exclude specific diagnoses (including iron overload, storage disorders, mitochondrial cytopathies, amyloidosis, and granulomatous myocardial diseases) and to diagnose restrictive myofibrillar disease caused by desmin variants.	IIa	C
ICD implantation may be considered in *children* with RCM who have evidence of myocardial ischaemia and syncope.[969]	IIb	C

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DCM, dilated cardiomyopathy; HCM, hypertrophic cardiomyopathy; ICD, implantable cardioverter defibrillator; PVR, pulmonary vascular resistance; RCM, restrictive cardiomyopathy.
[a]Class of recommendation.
[b]Level of evidence.

7.6. Syndromic and metabolic cardiomyopathies

It is beyond the scope of these guidelines to provide a detailed review and recommendations on specific cardiomyopathy genocopies and phenocopies. Instead, the Task Force refers the reader to detailed position statements and consensus documents published on behalf of the ESC Working Group on Myocardial and Pericardial Diseases (e.g. on Anderson–Fabry Disease and amyloidosis).[370,375,974] This section highlights only the key diagnostic and management issues. *Table 22* summarizes the clinical features and management of syndromic and metabolic cardiomyopathies.

7.6.1. Anderson–Fabry disease

7.6.1.1. Definition

Anderson–Fabry disease is an inborn error of metabolism where a deficient or absent enzyme, alpha-galactosidase A (α-Gal A), due to a pathogenic genetic variant in the *GLA* gene, causes the storage of some degradation cell products, mainly globotriaosylceramide (Gb3) in a patient's lysosomes.[975] This storage causes cell dysfunction in its own right and activates cellular hypertrophy pathways, common to other causes of HCM, as well as inflammation and immune activation.[976] It is a multisystem disorder affecting particularly the heart, kidney, and brain. [975] It is inherited in an X-linked manner; males are therefore always affected, while females' organ involvement usually develops later in life but can become similar to males due to the lyonization phenomena.[977,978]

Two Anderson–Fabry phenotypes can be distinguished, depending on the gender, lyonization phenomena, and pathogenic genetic variant:[976,979]

- A severe clinical phenotype, known as 'classic' Anderson–Fabry characterized by absent or severely reduced (<1% of mean normal) α-Gal A activity, marked Gb3 accumulation, and childhood or adolescent onset of symptoms followed by progressive multiorgan failure, is most often seen in males (but not exclusively) without residual enzyme activity.
- A 'non-classical' Anderson–Fabry phenotype or later-onset phenotype with incomplete systemic involvement, which is seen in both males and females, with some level of residual enzyme activity, and in most cases manifesting as isolated cardiac involvement.

7.6.1.2. Diagnosis, clinical work-up, and differential diagnosis

Anderson–Fabry disease should be suspected in patients with LVH and additional cardiac and extracardiac red flags (see *Table 23*) sought (*Figure 20*). The diagnosis is established by assessment of α-GalA activity and lyso-Gb3 measurement in male patients; in females, genetic testing is usually required to confirm the diagnosis. Severe LVH (>15 mm) is unlikely to be seen in patients <20 years of age.[980] In children and adolescents, diagnosis is made by family history or based on other extracardiac symptoms, but overt LVH is usually not present.[981]

7.6.1.3. Clinical course, outcome, and risk stratification

Cardiovascular involvement usually manifests as LVH, myocardial fibrosis, inflammation, heart failure, and arrhythmias, which limit QoL and are the most common cause of death. Clinical monitoring is essential to assess disease progression and requires a multidisciplinary approach.[980]

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Table 22 Clinical features and management of syndromic and metabolic cardiomyopathies

Clinical red flags	Diagnosis	Specific cause	Multidisciplinary team	Management
Abnormal facial features Cryptorchidism Pulmonary valve stenosis Congenital heart disease Extreme right-axis deviation at ECG Lymphangectasis Bleeding diathesis Café au lait spots Lentigines Growth retardation Sensorineural deafness	NGS panel testing for RASopathy	Noonan syndrome Costello syndrome Cardiofaciocutaneous syndrome Noonan syndrome with multiple lentigines	Cardiologist Geneticist Endocrinologist Paediatrician Dermatologist Radiologist	Beta-blockers/CCBs Selective management of RVOTO/ pulmonary valvuloplasty SCD risk stratification
Short PR interval End-stage, hypokynetic HCM AV block (Kearns–Sayre syndrome) Lactic acidosis Sensorineural deafness Neutropenia (Barth syndrome) Diabetes Stroke-like lesions at brain MRI	NGS panel for mtDNA and nuclear DNA Skeletal muscle biopsy/ endomyocardial biopsy	MELAS syndrome MERRF syndrome Leigh syndrome Other mitochondrial disease Beta-oxidation disorders	Cardiologist Neurologist Endocrinologist Paediatrician Metabolism expert Radiologist	Avoiding drugs or situational stressors Beta-oxidation disorders: nutritional management, avoidance of fasting, aggressive treatment during increased metabolic stress Carnitine supplementation (selected cases)
Hepatomegaly Increased aminotransferase enzymes Delayed motor milestones Hypotonia Short PR interval ECG criteria for extreme LVH	Screening: GAA activity in DBS or leucocytes/Glc_4 dosing Diagnostic confirmation: acid alpha-glucosidase assay performed on skin fibroblasts (preferred method) or muscle biopsy	Type II glycogen storage disease (Pompe disease)	Cardiologist General Paediatrician/ neonatologist Gastroenterologist Neuromuscular disease specialist	Enzyme replacement therapy
Short PR interval Massive LVH Skeletal myopathy Increased serum CK enzyme Intellectual disability X-linked inheritance	NGS or target testing for *LAMP-2* variants	Danon disease	Cardiologist Neuromuscular disease specialist Pneumologist Advanced heart failure specialist	No treatment
Short PR interval Early-onset atrial fibrillation AV block Increased serum CK enzyme Autosomal dominant inheritance pattern	NGS or target testing for *PRKAG2*	PRKAG2 syndrome	Cardiologist Neuromuscular disease expert	No treatment

Continued

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Progressive limb ataxia Diabetes mellitus Pes cavus Reduced native T1 at CMR imaging	NGS testing for bi-allelic expansion of GAA repeats in the *FXN* gene	Friedreich ataxia	Cardiologist Neurologist Endocrinologist Orthopaedic surgeon Neuromuscular disease expert	No specific treatment
Bilateral carpal tunnel syndrome Lumbar spinal stenosis Autonomic dysfunction Peripheral neuropathy Relative apical sparing pattern Ejection fraction/strain ratio >5 Pseudonecrosis Q waves Low ECG voltages OR Positive serum or urine monoclonal chain at immunofixation	DPD/HMDP Tc^{99} scintigraphy Free light chain/serum and urine immunofixation Endomyocardial biopsy	Cardiac amyloidosis (AL or ATTR) (see *Section 7.7*)	Cardiologist Neurologist Nephrologist Haematologist (AL amyloidosis) Ophthalmologist	Tafamidis Patisiran[a] Inotersen[a] (ATTR-CA) OR Specific chemotherapy (AL amyloidosis)
Gastrointestinal symptoms Angiokeratoma Cornea verticillata Chronic kidney disease Proteinuria Sensorineural hypoacusia Stroke/TIA Neuropathic pain X-linked inheritance pattern Short PR interval Low native T1 at cardiac CMR	Screening in males: lyso-Gb3 dosing Screening in females/diagnostic confirmation: genetic testing for *GLA* variants	Anderson–Fabry disease	Cardiologist Nephrologist Neurologist Ophthalmologist Audiologist Gastroenterologist Dermatologist	Enzyme replacement therapy (agalsidase alfa/beta) Migalastat
Skeletal myopathy Posterolateral pseudonecrosis pattern Posterolateral or inferolateral akinesia	Genetic testing for dystrophinopathies	*DMD*	Neurologist Cardiologist Pneumologist Neuromuscular disease expert	Steroids (prednisone or deflazacort)
Skeletal myopathy AV block Premature atrial fibrillation Malignant ventricular arrhythmias	NGS testing	*LMNA* cardiomyopathy Emery–Dreifuss muscular dystrophy	Cardiologist Neurologist	SCD risk prevention Pacing if indicated
Bilateral hilar lymphadenopathy Pulmonary infiltrates Uveitis Gastrointestinal involvement High-degree AV block Frequent VEs Thinned basal interventricular septum Extended LGE at CMR imaging	18F-FDG-PET Endomyocardial biopsy Lung biopsy	Sarcoidosis	Cardiologist Pneumologist Neurologist Gastroenterologist	Steroids Steroid-sparing immunosuppressant drugs

Continued

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Previous transfusions Chronic liver disease Skin pigmentation Diabetes Hypogonadotropic hypogonadism Elevated ferritin AV block	Iron status Complete blood count Increased T2* values at CMR imaging Genetic test for *HFE*, *HJV*, hepcidin receptor, ferroportin, *HAMP* gene Peripheral blood smear Haemoglobin electrophoresis Genetic testing for hereditary haemoglobinopathies	Iron overload cardiomyopathy	Cardiologist Haematologist Endocrinologist Paediatrician Gastroenterologist	Iron-chelating drugs Phlebotomy

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18F-FDG-PET, 18F-fluorodeoxyglucose positron emission tomography; AL, amyloid light chain; ATTR, transthyretin amyloidosis; ATTR-CA: transthyretin cardiac amyloidosis; AV, atrioventricular; CCB, calcium channel blocker; CK, creatinine kinase; CMR, cardiac magnetic resonance; DBS, deep brain stimulation; DMD, Duchenne muscular dystrophy; DPD, 3,3-diphosphono-1,2-propanodicarboxylic acid; ECG, electrocardiogram; Gb3, globotriaosylceramide; HCM, hypertrophic cardiomyopathy; HMDP, hydroxymethylene diphosphonate; LGE, late gadolinium enhancement; LMNA, lamin A/C; LVH, left ventricular hypertrophy; MELAS, mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes (syndrome); MERRF, mitochondrial epilepsy with ragged-red fibres; MRI, magnetic resonance; mtDNA, mitochondrial DNA; NGS, next-generation sequencing; PRKAG2, protein kinase AMP-activated non-catalytic subunit gamma 2; RVOTO, right ventricular outflow tract obstruction; SCD, sudden cardiac death; TIA, transient ischaemic attack; VE, ventricular ectopic beats.
[a]Patisiran and inotersen approved for treatment of familial polyneuropathy with/without cardiomyopathy.

Table 23 Anderson–Fabry disease red flags

Extracardiac red flags	
No male-to-male transmission in pedigree	
Renal involvement (dialysis, renal transplantation) or LVH in relatives	
Neuropathic pain	
Angiokeratomas	
Albuminuria	
Cornea verticillata	
Hypohidrosis, heat/cold and exercise intolerance	
Gastrointestinal symptoms (nausea, vomiting, non-specific abdominal pain, constipation, diarrhoea)	
Hearing loss (either progressive or sudden), tinnitus, vertigo	
Cardiac red flags	
ECG	Short PQ interval in young patients
	Atrioventricular blocks in adult patients
	Bradycardia
	Chronotropic incompetence
	LVH
Echocardiogram	LVH with normal systolic function
	Hypertrophy of papillary muscles
	Mitral and aortic valve thickening with mild-to-moderate regurgitation
	Reduced global longitudinal strain
CMR	Basal-inferolateral late gadolinium enhancement
	Low native T1 (caution with 'pseudonormalization' in areas affected by fibrosis)
	High focal/global T2
Laboratory	Elevated high-sensitivity troponin Elevated NT-proBNP

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CMR, cardiac magnetic resonance; ECG, electrocardiogram; LVH, left ventricular hypertrophy; NT-proBNP, N-terminal pro-brain natriuretic peptide.

7.6.1.4. Management

Specific treatment strategies, including enzyme replacement or pharmacological chaperone, have limited efficacy in advanced cases with irreversible organ damage, so early initiation appears to be important. Enzyme replacement therapy is indicated in all symptomatic patients with classical disease, including children, at the earliest signs of organ involvement.[974] Therapeutic strategies currently in development include second-generation ERTs, substrate-reduction therapies, and gene and mRNA therapies.[980]

7.6.2. RASopathies

7.6.2.1. Definition

The RASopathies constitute a group of multisystemic syndromes caused by variants in the RAS-mitogen-activated kinase (RAS-MAPK) cascade,[984–986] including Noonan syndrome,[987–989] Noonan syndrome with multiple lentigines;[990,991] Costello syndrome,[992,993] and cardiofaciocutaneous syndrome.[994–996]

7.6.2.2. Diagnosis, clinical work-up, and differential diagnosis

The suspicion of an underlying RASopathy should be raised in infant- and childhood-onset HCM with coexisting CHD[262,263,991,997–1000] or extracardiac abnormalities (see *Table 22*). Gene testing is recommended for diagnosis when phenotypic features are present. Compared with sarcomeric HCM, RASopathy-associated HCM (RAS-HCM) shows earlier age at diagnosis,[261,999] increased prevalence and severity of left or biventricular obstruction,[258,262,1001] and higher rates of early hospitalizations for heart failure or need for interventional procedures or surgery.[258] Pulmonary stenosis is the most commonly associated CHD, with a prevalence ranging between 25% and 70%, and unfavourable outcomes for pulmonary valvuloplasty.[256,1002–1004]

7.6.2.3. Clinical course, management, and sudden death risk stratification

Data from the North American Pediatric Cardiomyopathy Registry[1005] cohort show poorer survival rates among patients with RAS-HCM compared with non-syndromic HCM, particularly in patients who have been diagnosed before 1 year of age. Disease-specific risk factors for SCD are currently an area of debate, and may include the degree of LV hypertrophy, prolonged QTc interval, ECG risk score for HCM,[771] and the HCM Risk-Kids score >6%.[81,826]

7.6.2.4. Management

Non-vasodilating beta-blockers should be titrated to maximum tolerated dose in patients with RAS-HCM, particularly in cases of severe

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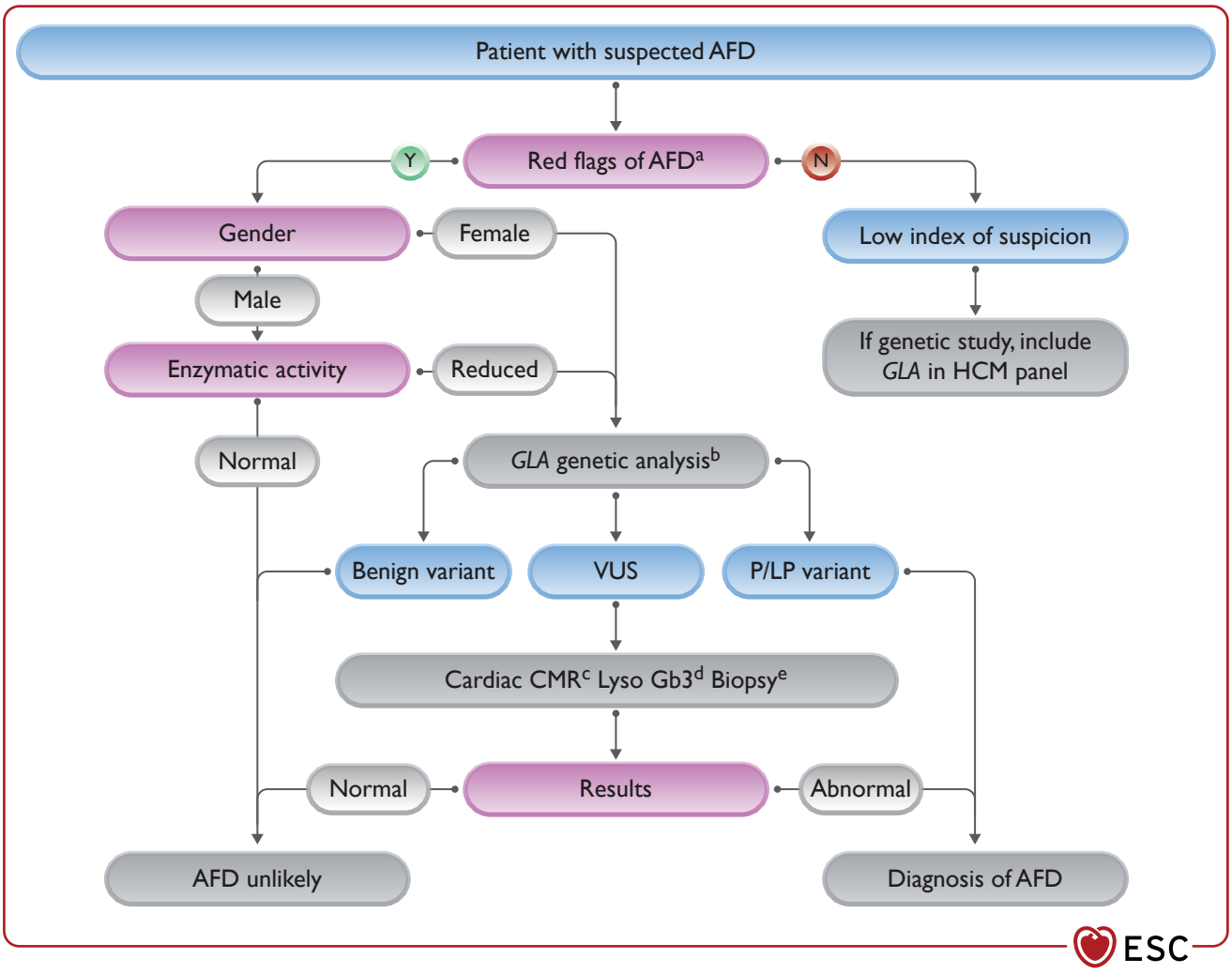


Figure 20 Anderson–Fabry disease diagnostic algorithm. α-Gal A, alpha-galactosidase A; AFD, Anderson–Fabry disease; CMR, cardiac magnetic resonance; Gb3, globotriaosylceramide; HCM, hypertrophic cardiomyopathy; LVH, left ventricular hypertrophy; lyso Gb3, globotriaosylsphingosine; P/LP, pathogenic/likely pathogenic; VUS, variant of unknown significance. [a]See *Table 23*. [b]Genetic analysis must include the study of possible large deletions or a copy number variation not detected by the Sanger method. [c]The finding of increased plasma and/or urinary Gb3, or plasma lyso Gb3 and its analogues in the evaluation of male or female patients with a VUS and normal (in female patients) or lowered α-Gal A activity provides additional diagnostic information, but the role of biomarkers in such patients still requires validation. [d]Low native T1 values reinforce or generate suspicion of Fabry disease. Normal native T1 values do not exclude Fabry disease, as they are rarely observed in untreated patients with mild LVH (mostly females), or in advanced disease due to pseudonormalization. [e]An endomyocardial biopsy is recommended, but could be done in other affected organs such as the kidneys and skin. It should be evaluated by expert pathologists and always include electron microscopy studies to detect lamellar bodies and intracellular inclusions. Of note, some drugs may produce drug-induced phospholipidosis with an intracellular accumulation of phospholipids in different organs that can mimic zebra bodies on electron microscopy.[982,983]

biventricular obstruction.[248,1002,1006–1008] Calcium channel blockers may be considered as a second-line option in patients >6 months of age when beta-blocker therapy is ineffective or not tolerated.[267,639] Surgical myectomy and orthotopic heart transplantation may be considered in high-volume centres after multidisciplinary assessment by the heart team.[265,266,1009–1011] Pulmonary valvuloplasty may be considered in children and infants with severe RV outflow tract obstruction (RVOTO).[1012–1015]

7.6.3. Friedreich ataxia

7.6.3.1. Definition

Friedreich ataxia is an autosomal recessive disorder caused by a homozygous GAA triplet repeat expansion in the frataxin (*FTX*) gene,[1016–1019] leading to HCM, progressive neuromuscular symptoms, and extracardiac manifestations, including diabetes mellitus.[1016,1020,1021]

7.6.3.2. Diagnosis, clinical work-up, and differential diagnosis

Although several diagnostic criteria have been proposed to suspect Friedreich ataxia,[1022,1023] genetic testing with identification of bi-co-allelic GAA expansion in the first intron of the *FTX* gene or compound heterozygosis is required for diagnosis.[1024,1025]

Cardiovascular involvement usually manifests as hypertrophic non-obstructive cardiomyopathy, with hypokinetic end-stage disease progression and impaired perfusion reserve,[1026] leading to advanced heart failure and death.[248,1005,1027–1029] There appears to be no specific relationship between the extent of neurological involvement and cardiac

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phenotype.[248,1005,1027–1029,1005,1027–1030,1030] Mitochondrial iron storage is the pathologic hallmark of the disease.[1031]

7.6.3.3. Clinical course, management, and risk stratification

Supraventricular arrhythmias, particularly AF, are commonly detected.[1027] Despite the lack of long-term follow-up longitudinal studies, the risk of ventricular arrhythmias and SCD seems low compared with sarcomeric HCM.[1027,1032,1033] The Mitochondrial Protection with Idebenone in Cardiac or Neurological Outcome (MICONOS) study group[1034] has proposed a staging of cardiac involvement based on LVEF and end-diastolic wall thickness. The extent of TWI at ECG, LVEF, LV end-diastolic posterior wall thickness, fibrosis on CMR, and hs-TnT have been proposed as negative prognostic factors.[1035]

7.6.3.4. Management

No specific treatment is currently available for Friedreich ataxia. Treatment with idebenone, a coenzyme Q_{10} analogue, showed the potential to improve LV mass and cardiac outcomes in open-label studies;[1036] nevertheless, four RCTs[1037–1040] showed no significant benefit on cardiac or neurologic outcomes.

7.6.4. Glycogen storage disorders

7.6.4.1. Definition

Glycogen storage disorders (GSDs) represent a heterogeneous group of metabolic diseases, including infantile-onset Pompe disease (GSD, type IIa), Danon disease (GSD, type IIb), and PRKAG2 disease.[272]

7.6.4.2. Diagnosis, clinical work-up, and differential diagnosis

Despite wide clinical heterogeneity, a presentation within the first few months of life, hypotonia, failure to thrive, generalized muscle weakness, and severe non-obstructive HCM with concentric pattern followed by hypokinetic end-stage cardiomyopathy, usually within the first year of life, are typical of GSD IIa.[259,268,1041,1042] Short PR interval and increased ECG voltages may represent useful diagnostic clues for GSDs.[1042,1043] PRKAG2 syndrome should be suspected in the setting of autosomal dominant transmission and association with conduction system disease including ventricular pre-excitation, sick sinus syndrome, AF, AV block, intraventricular conduction delays or sinoatrial blocks.[1043–1047] An X-linked pattern of inheritance is typical of Danon disease (GSD IIb). Skeletal myopathy, in association with learning disability, retinal involvement and ventricular pre-excitation, has been detected in males affected by Danon disease, while the cardiac phenotype can be isolated in affected females.[1048–1052]

7.6.4.3. Clinical course, management, and risk stratification

In the absence of therapeutic intervention, Pompe disease has a poor prognosis, mainly due to end-stage heart failure.[268,1041] Recently, data from a large multicentre European registry have shown that Danon disease runs a malignant phenotype, but there are insufficient data to identify candidate risk factors for sudden death.[1049] Sudden cardiac death occurs in almost 10% of patients with PRKAG2 syndrome, mainly as a consequence of advanced AV block, supraventricular tachycardia degenerated to VF, or massive hypertrophy.[1044,1053,1054]

7.6.4.4. Management

Enzyme replacement therapy is recommended in patients with GSD IIa.[269,274,275,1055,1056] To date, there are no approved aetiological therapies for PRKAG2 syndrome and Danon disease. Heart failure therapy, antiarrhythmic therapy, and indications for the implantation of devices are included in *Section 6.10*.

7.7. Amyloidosis

It is beyond the scope of this document to provide specific recommendations for the assessment and management of cardiac amyloidosis. Instead, the Task Force refers the reader to the 2021 position statement of the ESC Working Group on Myocardial and Pericardial Diseases on Diagnosis and Treatment of Cardiac Amyloidosis.[375] This section highlights only the key diagnostic and management issues.

7.7.1. Definition

Cardiac amyloidosis is characterized by the extracellular deposition of misfolded proteins in the ventricular myocardium with the pathognomonic histological property of green birefringence when viewed under cross-polarized light after staining with Congo Red.[375]

Although once considered a rare disease, data obtained in the last decade suggest that cardiac amyloidosis is underappreciated as a cause of common cardiac diseases or syndromes such as HFpEF, aortic stenosis, or unexplained LVH, particularly in the elderly.[1057–1059] Although nine different types of cardiac amyloidosis have been described, most cases correspond to monoclonal immunoglobulin light chain amyloidosis (AL) or transthyretin amyloidosis (ATTR), either in its hereditary (ATTRv) or acquired (ATTRwt) form.[375] The ATTRwt form, which is associated with ageing, is currently considered the most frequent form of cardiac amyloidosis worldwide.

7.7.2. Diagnosis, clinical work-up, and differential diagnosis

Cardiac amyloidosis should be suspected in patients with increased LV wall thickness in the presence of cardiac or extracardiac red flags and/or in specific clinical situations, as detailed in *Figure 21*, particularly in patients >65 years of age.[375]

Cardiac amyloidosis can be diagnosed using both invasive and non-invasive diagnostic criteria.[375] Invasive diagnostic criteria apply to all forms of cardiac amyloidosis, whereas non-invasive criteria are accepted only for ATTR. Invasive criteria include demonstration of amyloid fibrils within cardiac tissue or, alternatively, demonstration of amyloid deposits in an extracardiac biopsy accompanied either by characteristic features of cardiac amyloidosis on echocardiography or CMR.[375] Non-invasive criteria include typical echocardiographic/CMR findings combined with planar and single-photon emission computed tomography (SPECT) grade 2 or 3 myocardial radiotracer uptake in ^{99m}technetium-pyrophosphate (^{99m}Tc-PYP) or 3,3-diphosphono-1,2-propanodicarboxylic acid (DPD) or hydroxymethylene diphosphonate (HMDP) scintigraphy and exclusion of a clonal dyscrasia by all the following tests: serum free light chain assay, serum and urine protein electrophoresis with immunofixation.[168] Tomographic scintigraphy should be considered in order to reduce the number of misclassifications.[1060] False negative scans may rarely occur in certain ATTRv genotypes; false positives may be due to AL, recent myocardial infarction, or long-term chloroquine use.[370] Therefore, planar and SPECT scintigraphy coupled with assessment for monoclonal proteins followed by CMR and/or cardiac/extracardiac biopsy if necessary allows appropriate diagnosis in patients with suggestive signs/symptoms, as described in *Figure 22*.[375] However, the DPD/PYP/HMDP scan cannot distinguish between wild-type and mutated ATTR, and therefore TTR genetic testing is required. Of note, TTR genetic testing is recommended in all transthyretin amyloid cardiomyopathy (ATTR-CM) patients regardless of age, as 5% of

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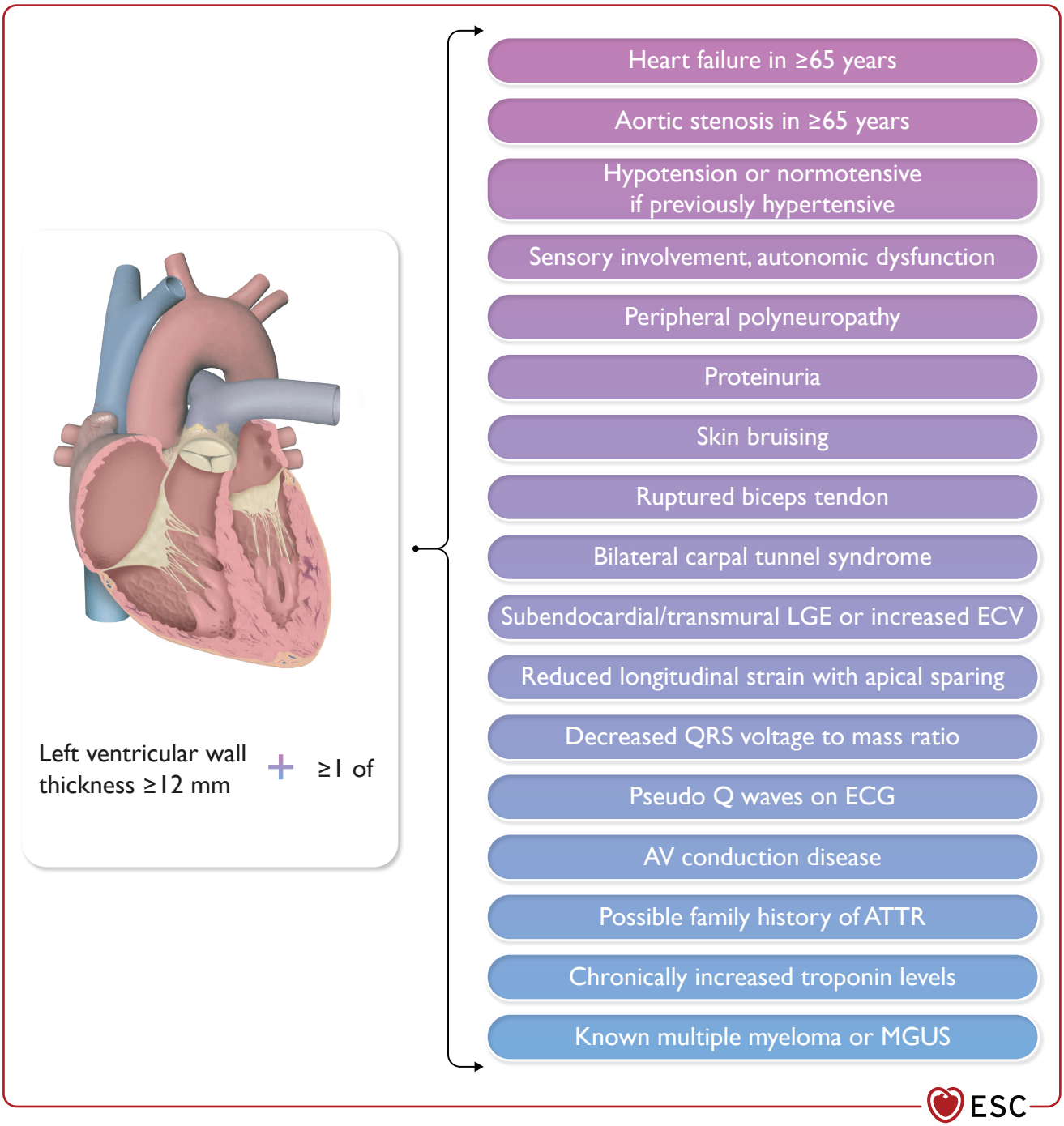


Figure 21 Screening for cardiac amyloidosis. ATTR, transthyretin amyloidosis; AV, atrioventricular; ECG, electrocardiogram; ECV, extracellular volume; LGE, late gadolinium enhancement; MGUS, monoclonal gammopathy of undetermined significance.

ATTR-CM patients ≥70 years (and 10% among females) have ATTRv.[375,1061]

7.7.3. Clinical course and risk stratification

Cardiac amyloidosis is a progressive disease with poor outcomes if left untreated. Amyloid light chain cardiac amyloidosis is associated with more rapid progression of heart failure and worse prognosis than ATTR.[1058,1062,1063] Fortunately, the prognosis of AL amyloidosis has significantly improved with the introduction of very effective therapies capable of dramatically reducing the production of the cardiotoxic light chains.[1064] Prognosis in ATTR depends on the variant, degree of cardiac involvement, and neurologic phenotype.[1065–1068] Several multiparametric biomarker-based staging systems have been developed for AL[1069,1070] and ATTR cardiac amyloidosis[1066–1068] (see Supplementary data online, *Table S5*).

7.7.4. Management

The treatment of cardiac amyloidosis includes treating and preventing complications and stopping or delaying amyloid deposition by specific treatment.[375,1071] There is no evidence to support the use of standard

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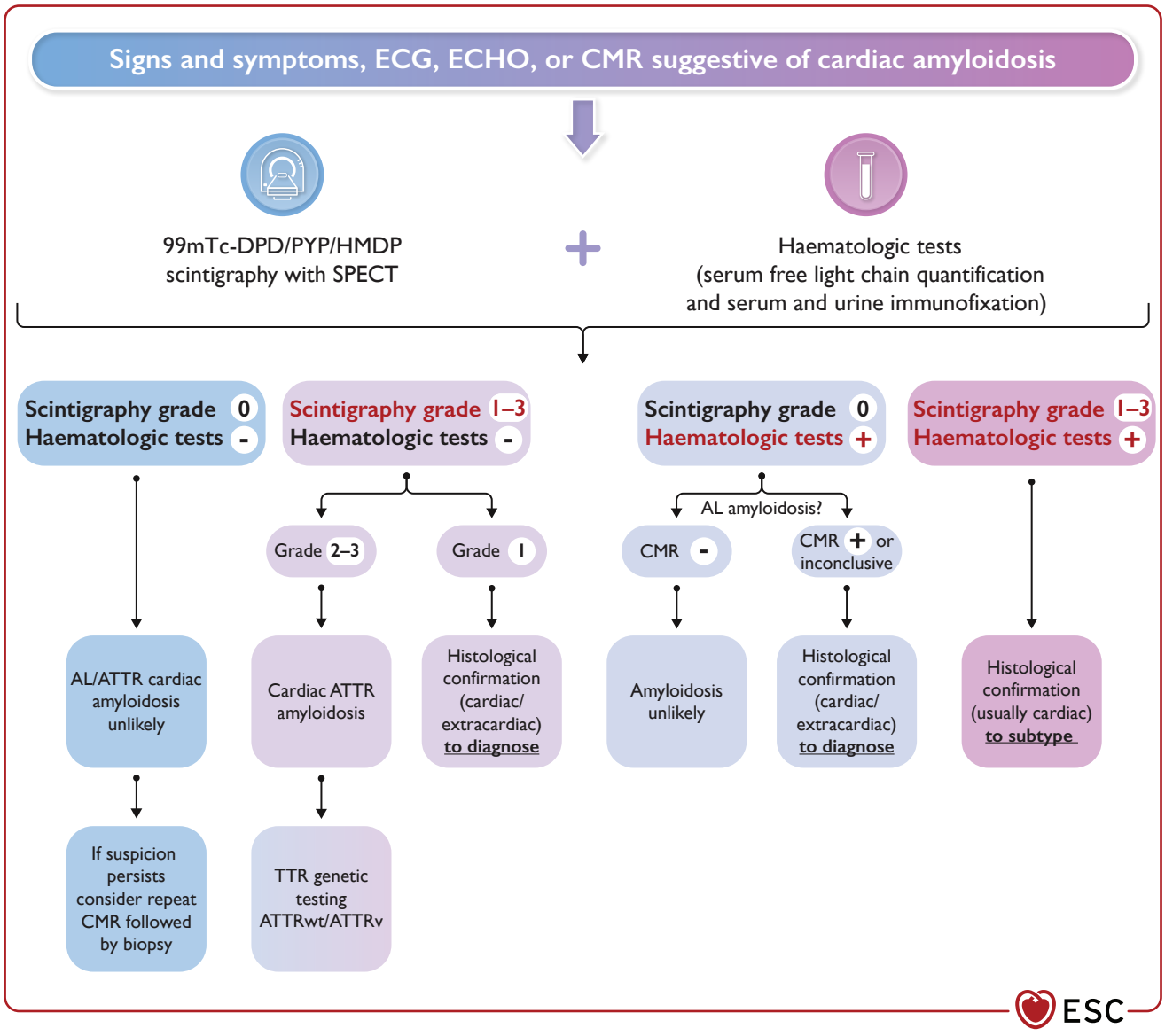


Figure 22 Diagnosis of cardiac amyloidosis. AL, amyloid light chain; ATTR, transthyretin amyloidosis; ATTRv, variant transthyretin amyloidosis; ATTRwt, wild-type transthyretin amyloidosis; CMR, cardiac magnetic resonance; DPD, 3,3-diphosphono-1,2-propanodicarboxylic acid; ECG, electrocardiogram; ECHO, echocardiogram; HMDP, hydroxymethylene diphosphonate; PYP, pyrophosphate; TTR, transthyretin.

heart failure therapy, which often is not well tolerated, apart from diuretics (see *Section 6.10.2*).[1072,1073]

The natural history of cardiac amyloidosis associates electrical conduction disease of the heart with symptomatic bradycardia and advanced AV block.[375,1074,1075] The clinical threshold for pacemaker indication should be low, as the disease progresses and implantation of the device allows rate response to exercise and medication adjustment.[375,1074] The role of ICD in cardiac amyloidosis for SCD prevention is not clearly known, but available data do not support their use in primary prevention.[1076,1077]

7.7.4.1. Specific therapies

Therapy for AL cardiac amyloidosis is based on treatment of the underlying haematological problem with chemotherapy or autologous stem-cell transplant.[1064]

Transthyretin stabilization and reduction of its production are the basis of TTR cardiac amyloidosis treatment. Tafamidis reduced all-cause mortality and cardiovascular hospitalizations in ATTR, with the largest effect achieved in patients at NYHA functional class I and II.[1078] Additional studies are being conducted with other stabilizing agents and other molecules that reduce TTR production.[1078a]

8. Other recommendations

8.1. Sports

8.1.1. Cardiovascular benefits of exercise

Regular physical activity and systematic exercise confer several cardiovascular, psychological, and QoL benefits. Through curbing risk factors for atherosclerosis, such as obesity and insulin resistance,[1079] hypertension,[1080] and hyperlipidaemia,[1081] regular physical activity is associated with an up to 50% risk reduction in an adverse event from CAD in middle-aged and older individuals.[1082,1083] Individuals who exercise regularly live 5–7 years longer than their sedentary counterparts,[1084]

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and have a lower risk of cerebrovascular accidents[1085] and certain malignancies.[1085–1087] These benefits that can be derived later in life also apply to individuals with established cardiovascular disease. For a definition of exercise intensity levels, please refer to Supplementary data online, *Table S6*.

8.1.2. Exercise-related sudden cardiac death and historical exercise recommendations for patients with cardiomyopathy

Rigorous exercise may trigger myocardial infarction and fatal arrhythmias among individuals with an underlying cardiovascular disease.[1088–1091] Superimposed on the pathological substrate of the disease entity itself, exercise may induce sudden cardiac arrest through mechanical shearing forces within the coronary arteries, effects of high concentrations of circulating catecholamines, increased cardiac loading conditions, raised core temperature, electrolyte shifts, and acid-base disturbance.

Cardiomyopathies are the leading cause of exercise-related SCD in young people in the Western world.[40,1092–1095] The established link between exercise and SCD from cardiomyopathy, and the finding that, in some cardiomyopathy phenotypes, exercise may accelerate progression of the underlying cardiomyopathic disease process, has historically resulted in restrictive exercise recommendations in all affected patients regardless of pathology, disease severity, symptomatic status, general risk profile, or prior therapeutic interventions, including an ICD.[1096–1098] As a result, individuals with cardiomyopathy often confine themselves to a relatively sedentary lifestyle through fear of potential SCD and accrue risk factors for atherosclerotic CAD, which confer a worse prognosis.[1099–1102,1096,1097]

8.1.3. Exercise recommendations in hypertrophic cardiomyopathy

Recent pre-clinical[1103] and clinical data suggest that moderate exercise may be beneficial and safe in patients with HCM.[1098–1102] Information on a safe dose of vigorous exercise is still limited, but the heterogeneous morphology and pathophysiology of HCM means that some individuals are capable of participating in vigorous exercise, including high-intensity competitive sports.[760] Most athletes capable of exercising intensively have mild LV hypertrophy, normal-sized or enlarged LV, normal diastolic function, and no evidence of LVOTO.[1104,1105] Currently available data indicate that participation in vigorous exercise and competitive sport may be considered in a select group of predominantly adult patients who have mild morphology and a low-risk profile.[1106–1108] However, studies examining the effect of vigorous exercise or moderate-to-high-intensity competitive sport on the natural history of HCM were not designed or powered adequately to address the question and there are potential issues of selection bias. Nevertheless, based on emerging evidence, the Task Force agreed to adopt a comparatively liberal approach, advocating that, after appropriate selection, some individuals with a low-risk profile may participate in high-intensity exercise and competitive sport after comprehensive expert evaluation and shared discussion, which highlights the unpredictable nature of exercise-related SCD in HCM. Sporting disciplines in which syncope may result in fatal accidental injury or danger to others are not recommended.

Genotype-positive/phenotype-negative patients may engage in all competitive sport; however, annual assessment is recommended to check for developing phenotypic features of disease.[1109]

8.1.4. Exercise recommendations in arrhythmogenic right ventricular cardiomyopathy

Arrhythmogenic right ventricular cardiomyopathy is a recognized cause of exercise-related SCD in young asymptomatic individuals,[40,890] postulated to result from ventricular stretch leading to myocyte detachment with subsequent inflammation and fibro-fatty replacement of the ventricular myocardium. Fatal arrhythmias may occur during the inflammatory process or because of myocardial scar. In addition, there are data to suggest that high-intensity exercise is associated with acceleration of disease phenotype in individuals with ARVC, including those who are genotype positive/phenotype negative, and particularly those with *PKP2* variants.[181,1110–1114] Furthermore, exercise restriction has been shown to improve clinical outcomes in patients with ARVC.[40,1111,1115–1117] Based on these data, the Task Force recommends against intensive exercise or competitive sports in individuals with ARVC as part of a shared decision-making process. The evidence on the impact of exercise in genotype-positive/phenotype-negative individuals is more limited. In these cases, the Task Force recommends a cautious approach in the context of shared decision-making when discussing competitive sports participation. Mild-to-moderate physical activity for up to 150 min per week is considered safe and is recommended in able phenotype-negative individuals.[1118]

8.1.5. Exercise recommendations in dilated cardiomyopathy and non-dilated left ventricular cardiomyopathy

There is evidence that moderate exercise in optimally treated patients with DCM improves functional capacity, ventricular function, and QoL;[1119] however, intensive exercise and competitive sports may also trigger fatal arrhythmias in DCM and NDLVC.[1093,1120–1122]

In general, symptomatic individuals with DCM and NDLVC should abstain from most competitive and leisure sports, or recreational exercise associated with moderate or high exercise intensity. A select group of asymptomatic individuals with DCM and NDLVC who have mildly impaired LV function without exercise-induced arrhythmias or significant myocardial fibrosis may participate in most competitive sports.

Although the natural history of most pathogenic variants capable of causing DCM and NDLVC is unknown, it would be reasonable to permit intensive exercise and competitive sports in most individuals with pathogenic variants in the absence of overt features of DCM or NDLVC. Special consideration, however, should be given to individuals with pathogenic variants in genes that are associated with an increased risk of life-threatening arrhythmias, such as lamin A/C[181,1123] or *TMEM43* variants, for which there is emerging evidence that exercise may have an adverse effect on cardiac function and risk of potentially fatal arrhythmias. The impact of vigorous exercise in patients with pathogenic variants in other high-risk genes, such as filamin C variants[1112] exhibiting DCM or NDLVC phenotypes, is not fully understood; however, extrapolating our understanding of the effect of exercise on some ARVC and DCM phenotypes necessitates a cautious approach.

Recommendation Table 31 — Exercise recommendations for patients with cardiomyopathy

Recommendations	Class[a]	Level[b]
All cardiomyopathies		
Regular low- to moderate-intensity exercise is recommended in all able individuals with cardiomyopathy.	I	C

Continued

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An individualized risk assessment for exercise prescription is recommended in all patients with cardiomyopathy.	I	C
HCM		
High-intensity exercise and competitive sport should be considered in genotype-positive/phenotype-negative individuals who seek to do so.[1124]	IIa	C
High-intensity exercise and competitive sport may be considered in asymptomatic low-risk[c] individuals with morphologically mild hypertrophic cardiomyopathy in the absence of resting or inducible left ventricular outflow obstruction and exercise-induced complex ventricular arrhythmias.[1107,1113,1125,1126]	IIb	B
High-intensity exercise, including competitive sport, is not recommended in high-risk individuals and in individuals with left ventricular outflow tract obstruction and exercise-induced complex ventricular arrhythmias.	III	C
ARVC		
Avoidance of high-intensity exercise, including competitive sport, may be considered in genotype-positive/phenotype-negative individuals in families with ARVC.[1111,1116,1117]	IIb	C
Moderate- and/or high-intensity exercise, including competitive sport, is not recommended in individuals with ARVC.[181,1111–1114]	III	B
DCM and NDLVC		
Moderate- and high-intensity exercise should be considered in individuals who are gene positive and phenotype negative (with the exception of pathogenic variants in *LMNA* and *TMEM43*) who seek to do so.[1123]	IIa	C
High-intensity exercise and competitive sport may be considered in a select group of asymptomatic and optimally treated individuals with a left ventricular ejection fraction ≥50% in the absence of exercise-induced complex arrhythmias.	IIb	C
Moderate-intensity exercise may be considered in asymptomatic and optimally treated individuals with a left ventricular ejection fraction of 40–49% in the absence of exercise-induced complex arrhythmias.	IIb	C
High-intensity exercise, including competitive sport, is not recommended in symptomatic individuals, those with a left ventricular ejection fraction ≤40%, exercise-induced arrhythmias or pathogenic variants in *LMNA* or *TMEM43*.	III	C

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ARVC, arrhythmogenic right ventricular cardiomyopathy; DCM, dilated cardiomyopathy; HCM, hypertrophic cardiomyopathy; LMNA, lamin A/C; NDLVC, non-dilated left ventricular cardiomyopathy; TMEM43, transmembrane protein 43.
[a]Class of recommendation.
[b]Level of evidence.
[c]See *Section 7.1.5* for risk assessment in HCM.

8.2. Reproductive issues

Pregnancy and the post-partum period constitute periods of increased risk of cardiovascular complications in women with cardiomyopathy.[1127–1130] Cardiomyopathy can also be first diagnosed in pregnancy or arise during pregnancy as PPCM.[1131]

The risk associated with pregnancy in a patient with a cardiomyopathy is estimated using the modified World Health Organization (mWHO) classification.[1130] Pregnancy is contraindicated in women with WHO class IV, including patients with EF <30% or NYHA class III–IV or previous PPCM with persisting impairment of the LV function.

8.2.1. Contraception, *in vitro* fertilization, and hormonal treatment

Counselling on safe and effective contraception is indicated in all women of fertile age. Ethinyloestradiol-containing contraceptives have the greatest risk of thrombosis[1132] and are not advised in women with a high risk of thrombo-embolic disease. Progestin-only contraceptives are an alternative, as they have little or no effect on coagulation factors, blood pressure, and lipid levels. Levonorgestrel-based long-acting reversible contraception implants or intrauterine devices are the safest and most effective contraceptives and have few side effects affecting cardiomyopathies.

Medically assisted procreation adds risks beyond those of pregnancy alone; superovulation is pro-thrombotic and can be complicated by ovarian hyperstimulation syndrome, with marked fluid shifts and an even greater risk of thrombosis. Hormonal stimulation should be carefully considered in women who have WHO class III disease (VT or HCM) or who are anticoagulated.

8.2.2. Pregnancy management

8.2.2.1. Pre-pregnancy

Patients with a known cardiomyopathy and at risk of developing cardiomyopathy should receive pre-pregnancy counselling by a multidisciplinary team: the pregnancy heart team. The individual risk of the woman by pregnancy should be discussed using the WHO classification, in addition to discussing the likelihood of transmission of the disease to the offspring and how to reduce the transgenerational risk of transmitting the disorder.

For individual risk estimation, at a minimum, an ECG, echocardiography, and exercise test should be performed. Several aspects must be discussed with the woman, including long-term prognosis, drug therapy, estimated maternal risk and outcome, and plans for pregnancy care and delivery.

8.2.2.2. Pregnancy

In women with mWHO class II–III, III, and IV (including women with HCM, VTs, and EF <35%), management during pregnancy and around delivery should be conducted in an expert centre by a multidisciplinary team: the pregnancy heart team, including cardiologists with expertise in cardiomyopathies and arrhythmias; obstetricians; and anaesthetists. Depending on the individual case, other specialists may be included (geneticist, cardio-thoracic surgeon, paediatric cardiologist, foetal medicine specialist, neonatologist, etc.). A delivery plan should be created that includes the details of induction; the management of labour and delivery; and post-partum surveillance.

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8.2.2.3. Timing and mode of delivery

The timing and mode of delivery should be personalized according to the type of cardiomyopathy, ventricular function, NYHA class, arrhythmic risk, and thrombo-embolic risk. Vaginal delivery is associated with less blood loss and lower risk of infection, venous thrombosis, and embolism than caesarean section and should be advised for most women. Caesarean section should be considered for obstetric indications, patients with severe outflow tract obstruction, or in cases of severe acute/intractable heart failure, or in cases at high risk of threatening arrhythmia and for patients presenting in labour on oral anticoagulants.[1130] During delivery, patients with cardiomyopathy should be circulatory and heart rhythm monitored on an individualized basis.

8.2.2.4. Post-partum

The post-partum period is associated with significant haemodynamic changes and fluid shifts, particularly in the first 24–48 h after delivery, which may precipitate heart failure. Haemodynamic monitoring should therefore be continued for at least 24–48 h in patients at risk. Most drugs enter the milk and could thus contraindicate breastfeeding (see *Section 8.2.2.5*).

8.2.2.5. Pharmacological treatment: general aspects

Pharmacological treatment in pregnant women should be the same as in non-pregnant patients, with an avoidance of drugs contraindicated in pregnancy, such as ACE-Is, ARBs, and renin inhibitors.[1130] The first trimester is associated with the greatest teratogenic risk. Pharmacologic therapy is advised to begin as late as possible in pregnancy and at the lowest effective dose. Drug exposure later in pregnancy may confer adverse effects on foetal growth and development. It is recommended to check drug and safety data before initiation of a new drug in pregnancy; see Table 7 in the *2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy*.[1130] From this list, antiarrhythmics can be summarized as follows:

- Well tolerated: sotalol, oral verapamil.
- While the benefits and risks should be evaluated in each case, the following drugs can often be continued if there is a clear indication for use during pregnancy: bisoprolol, carvedilol, digoxin, diltiazem (possible teratogenic effects), disopyramide (uterine contractions), flecainide, lidocaine, metoprolol, nadolol, propranolol, verapamil, quinidine.
- Insufficient data: ivabradine, mexiletine, propafenone, vernakalant.
- Contraindicated: amiodarone, atenolol, dronedarone, ACE-Is, ARBs, renin inhibitors, and spironolactone.[1130]

Ongoing beta-blocker treatment in cardiomyopathies should be continued during pregnancy, with close monitoring of foetal growth. After delivery, it is advised to heart rhythm monitor the infant for 48 h. The use of beta-blockers and anticoagulation during pregnancy is described in the *2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy*.[1130]

Vitamin K antagonist use in the first trimester results in embryopathy (limb defects, nasal hypoplasia) in 0.6–10% of cases.[1133,1134] In contrast, unfractionated heparin (UFH) and low-molecular-weight heparin (LMWH) do not cross the placenta; therefore, substitution of VKA with UFH or LMWH in weeks 6–12 almost eliminates the risk of embryopathy. This risk is also dose dependent (0.45–0.9% with low-dose warfarin). Vaginal delivery while the mother is on VKAs is contraindicated because of the risk of foetal intracranial bleeding. Haemorrhagic complications in the mother occur with all regimens, but the incidence is lower with VKA than with LMWH/UFH throughout pregnancy.[1130]

VKA should be continued until pregnancy is achieved. Continuation of VKAs throughout pregnancy should be considered when the dose is low (see Table 7 in the *2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy*[1130]). The target international normalized ratio (INR) should be chosen according to current guidelines, with INR monitoring weekly or every 2 weeks. Self-monitoring of INR in suitable patients is recommended. Alternatively, depending on the indication, a switch to LMWH from weeks 6–12 under strict monitoring may be considered in patients with a low dose requirement. When a higher dose of VKAs is required, discontinuation of VKAs between weeks 6 and 12 and replacement with adjusted-dose i.v. UFH or LMWH twice daily with dose adjustment according to peak anti-Xa (for LMWH) levels should be considered.

In case of delivery in anticoagulated women (not including mechanical valves) with a planned caesarean section, therapeutic LMWH dosing can be simply omitted for 24 h prior to surgery. If delivery has to be performed earlier, anti-Xa activity can guide the timing of the procedure.

Antiarrhythmic therapy in pregnancy other than medication. Implantation of an ICD and catheter ablation should ideally be considered prior to pregnancy in patients with a high risk of ventricular arrhythmias to avoid implantations and interventions during pregnancy.[1135] If an ICD is indicated in pregnancy, ICD implantation should be performed beyond 8 weeks of gestation with radiation protection[1136] and the indication should be weighed against the limited experience available. In pregnant patients with existing ICD, routine ICD interrogation and advice are recommended prior to delivery.

8.2.2.6. Specific cardiomyopathies

Most women with HCM tolerate pregnancy well.[1137] Complications during pregnancy most often occur in women who have symptoms, arrhythmias, or impaired LV function before pregnancy. Left ventricular outflow tract gradients may increase slightly during pregnancy and high gradients before pregnancy are associated with more complications.[1137] Women should be assessed according to WHO risk class, indicating at trimester for low-risk patients (class II) and monthly or bi-monthly for higher-risk patients (class III). Therapeutic anticoagulation with LMWH or VKAs according to the stage of pregnancy is recommended for patients with AF. Cardioversion in pregnancy should be considered for poorly tolerated persistent AF. Hypovolaemia is poorly tolerated. Caesarean section should be considered in patients with severe LVOTO, pre-term labour while on oral anticoagulants, or severe heart failure.[1130] Epidural and spinal anaesthesia must be applied cautiously, especially with severe LVOTO, due to potential hypovolaemia, and single-shot spinal anaesthesia should be avoided.

Pregnancy in ARVC seems to be relatively tolerable, as shown in several studies, with no excess mortality and no clear negative long-term outcome.[1138–1141] Previous VTs represent a WHO risk class III, indicating bi-monthly or monthly follow-up at an expert centre.

Women with DCM are at risk of further deterioration of LV function in pregnancy. Data suggest that pregnancy might not be associated with long-term adverse disease progression or event-free survival in LMNA genotype-positive women.[1142] Predictors of maternal mortality are NYHA class III/IV and EF <40%. Highly adverse risk factors include

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EF <20%, severe mitral regurgitation, RV failure, AF, and/or hypotension.[1143]

8.2.2.7. Peripartum cardiomyopathy

Genetic studies in patients with PPCM have revealed genetic similarity between PPCM and DCM. Specifically, an overrepresentation of truncating variants has been demonstrated in *TTN*, *FLNC*, *BAG3*, and *DSP*, with *TTN* truncating variants most commonly involved (found in ~10% of patients).[44,45] It has been suggested that approaches to genetic testing in PPCM should mirror those taken in DCM.[45] Medications used to treat heart failure during pregnancy require special considerations as discussed above. In the presence of persistant cardiac dysfunction, medication should be continued. Use of bromocriptine as disease-specific therapy in patients with PPCM as an addition to standard heart failure therapy has shown promising results in two clinical trials.[1144,1145] In severe cases of PPCM, temporary MCS has been used successfully and should be considered in patients with haemodynamic instability despite inotropic support.[1146] In patients with PPCM, thresholds for early ICD implantations should be higher than in other conditions because of a high rate of spontaneous recovery after delivery.[1147]

Recommendation Table 32 — Recommendations for reproductive issues in patients with cardiomyopathy

Recommendations	Class[a]	Level[b]
Pre-pregnancy risk assessment and counselling are recommended in all women using the mWHO classification of maternal risk.	I	C
Counselling on safe and effective contraception is recommended in all women of fertile age and their partners.	I	C
Counselling on the risk of disease inheritance is recommended for all men and women before conception.	I	C
Vaginal delivery is recommended in most women with cardiomyopathies, unless there are obstetric indications for caesarean section, severe heart failure (EF <30% or NYHA class III–IV), or severe outflow tract obstructions, or in women presenting in labour on oral anticoagulants.	I	C
It is recommended that medication be carefully reviewed for safety in advance of pregnancy and adjusted according to tolerability in pregnancy.	I	C
Therapeutic anticoagulation with LMWH or VKAs according to the stage of pregnancy is recommended for patients with AF.	I	C
Continuation of beta-blockers should be considered during pregnancy in women with cardiomyopathies, with close follow-up of foetal growth and of the condition of the neonate, and if benefits outweigh risks.	IIa	C
Genetic counselling and testing should be considered in patients with peripartum cardiomyopathy.	IIa	C

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AF, atrial fibrillation; EF, ejection fraction; LMWH, low-molecular-weight heparin; mWHO, modified World Health Organization; NYHA, New York Heart Association; VKA, vitamin K antagonist.
[a]Class of recommendation.
[b]Level of evidence.

8.3. Recommendations for non-cardiac surgery

Cardiomyopathies, in general, are associated with an increased incidence of peri-operative heart failure and arrhythmias, although the significant variability in the phenotypic expression of cardiomyopathies must be considered. Special attention should be given to the clinical status, LVEF, volume overload, and increased levels of natriuretic peptides. In the period after non-cardiac surgery (NCS), fluids given during the operation may be mobilized, causing hypervolaemia and pulmonary congestion. Careful attention to fluid balance is therefore essential.[1148,1149] Obstructive HCM deserves specific consideration due to its peculiar pathophysiology, with adequate intra-operative vigilance, avoiding factors and medication that may increase LVOTO and prompt pharmacological treatment and intravascular fluid therapy if needed (see Supplementary data online, *Table S7*).[1150,1151]

Natriuretic peptide concentrations are quantitative plasma biomarkers for the presence and severity of haemodynamic cardiac stress and heart failure, and elevated NT-proBNP concentrations may facilitate detection of heart failure, optimal intra-operative monitoring, and initiation or optimization of heart failure therapy after surgery.[1152] Moreover, in cardiomyopathy patients elevated NT-proBNP values are strong predictors of overall prognosis.[1153–1156]

Patients with a first-degree relative with a genetic cardiomyopathy should be evaluated with an ECG and an echocardiographic examination to rule out the presence of the disease, irrespective of age (see *Section 6.11*). There are no specific data on risks of NCS in phenotype-negative family members; however, they are at risk of developing the disease, which may be subclinical at the time of the NCS.[1157] Data in children with HCM undergoing general anaesthesia for cardiac and non-cardiac procedures show that, in a specialist setting with multidisciplinary involvement, peri-operative morbidity and mortality are extremely low.[1158]

Recommendation Table 33 — Recommendations for non-cardiac surgery in patients with cardiomyopathy

Recommendations	Class[a]	Level[b]
Peri-operative ECG monitoring is recommended for all patients with cardiomyopathy undergoing surgery.	I	C
In patients with cardiomyopathy and suspected or known HF scheduled for intermediate or high-risk NCS, it is recommended to re-evaluate LV function with echocardiography (assessing LVOTO in HCM patients) and measurement of NT-proBNP/BNP levels, unless this has recently been performed.[1151,1153–1156,1158–1165]	I	B
It is recommended that cardiomyopathy patients with high-risk genotypes or associated factors for arrhythmic or heart failure complications or severe LVOTO be referred for additional specialized investigations to a cardiomyopathy unit before undergoing elective NCS.	I	C
In patients aged <65 years with a first-degree relative with a cardiomyopathy, it is recommended to perform an ECG and TTE before NCS, regardless of symptoms.	I	C

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ECG, electrocardiogram; HCM, hypertrophic cardiomyopathy; HF, heart failure; LV, left ventricular; LVOTO, left ventricular outflow tract obstruction; NCS, non-cardiac surgery; NT-proBNP, N-terminal pro-brain natriuretic peptide; TTE, transthoracic echocardiography.
[a]Class of recommendation.
[b]Level of evidence.

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9. Requirements for specialized cardiomyopathy units

As genomic tests and information are incorporated into strategies for the routine diagnosis and management of cardiomyopathies and the estimation of disease risk, cardiologists need to familiarize themselves with the general principles underlying the interpretation of test results and must be able to convey the implications to patients. They also need to be able to make informed judgments about which tests are appropriate for different patients and clinical situations. The risk of SCD and the possibility that family members could inherit the condition makes multidisciplinary expertise, including genetic counselling, psychological care, and patient support associations, a critical aspect of care.[1166] As a result, there is a growing need for clinicians to develop their understanding of the basic principles of clinical genetics and the diverse clinical manifestations of individual genetic disorders.[54,964,1166,1167]

Cardiomyopathies have a highly heterogeneous clinical presentation and an evolution that sometimes is difficult to predict. Disease phenotype can be the result of various acquired factors or genetic backgrounds. Mixed phenotypes or two conditions within the same patient or among a family can coexist. Genetic diagnosis raises common logistical and ethical problems in its execution, as well as in the interpretation and communication of the results.[1166] Diagnostic process, the management of symptoms, and risk stratification often require comprehensive evaluation of the patient and their family, with the participation of multidisciplinary teams. On the other hand, interventional procedures (septal ablation, myectomies, etc.) require an expertise that only centres that treat many patients can achieve. Specialization in this area also requires permanent updating to accurately characterize the disease prognosis, ensure the choice of the best therapeutic option in each case, and guarantee the implementation of that choice by a team with experience in the field.

These characteristics imply that the adequate management of these diseases requires specific tools, extensive experience, and a multidisciplinary basic-clinical approach that are difficult to achieve.

The cardiomyopathy unit is usually integrated into a general cardiogenetic (or inherited cardiac conditions) unit, where other professionals involved in hereditary cardiac and vascular conditions, such as channelopathies, genetic aortopathies, familial dyslipidaemias, and a number of genetic metabolic and syndromic diseases with cardiac involvement, are co-ordinated. They represent an organizational model aimed at providing comprehensive cardiovascular and genetic assessment and personalized management in patients with inherited cardiovascular diseases. Specialized multidisciplinary clinics have long been advocated as the ideal model for the management of patients and families with inherited cardiac conditions.[4,53,559,1166] Such a model of care supports the holistic care of patients and their at-risk family members, taking a patient-centred approach and valuing clinical, genetic, and psychosocial outcomes. The benefit of a specialized clinic for management of HCM has been previously reported, with patients showing better adjustment and less worry than those who did not attend.[53,224] Besides expertise in the field of inherited cardiac conditions, the presence of a multidisciplinary team, access to good technical resources, participation in dedicated research projects, availability of genetic counselling, and family screening are all pre-requisites for organizing a cardiogenetic clinic. The ability to provide education and training for medical professionals and collaboration with patients' associations is of utmost importance.

Supplementary data online, *Table S8* synthesizes the requirements and skills and recommendations for professional education/training needed for a cardiogenetic clinic as proposed by international expert associations.

10. Living with cardiomyopathy: advice for patients

Most people with cardiomyopathy lead normal and productive lives, but a small number experience significant symptoms and are at risk of disease-related complications. Regardless of the severity of their disease, it is important that individuals receive support and accurate advice from cardiomyopathy specialists and other healthcare professionals, and that they are encouraged to understand and manage the disease themselves (see the Supplemental Data online, *Table S9*, for a description of the patient education process).

Table 24 General guidance for daily activity for patients with cardiomyopathies

Topic	General guidance
Exercise	• See earlier section on exercise recommendations.
Diet, alcohol use, and weight	• Patients should be encouraged to maintain a recommended body mass index. • Avoid dehydration, excess alcohol intake, and drugs consumption.
Smoking	• There are no data that show an interaction between tobacco smoking and cardiomyopathy, but patients should be provided with general advice on the health risks associated with smoking, including pro-arrhythmic and pro-inflammatory effects and, when available, information on smoking cessation.[1168–1171]
Reproductive issues	• Patients should be given the opportunity to discuss their concerns about reproductive issues. Anxiety and depression following a diagnosis are frequent, and some patients may express guilt or fear about their genetic diagnosis and the risk of transmission to offspring.
Sexual activity	• Patients should be counselled on the potential effect of their medication on sexual performance. • Most people with cardiomyopathy will be able to undertake normal sexual activity. Individualized advice should be provided regarding its safety and the possible impact of sexual activity on the risk of disease progression, ventricular arrhythmias, and/or ICD shocks.
Medication	• Patients should be provided with information about their medication, including potential side and teratogenic effects and interactions with prescribed medications, over-the-counter remedies, and other complementary therapies.
Vaccination	• In the absence of contraindications, patients should be advised to have regular recommended vaccinations (e.g. yearly influenza and SARS-CoV-2 vaccination).

Continued

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Driving	• Most patients should be eligible for an ordinary driving licence and can continue driving unless they experience distracting or disabling symptoms. • Advice on driving licences for heavy goods or passenger-carrying vehicles should be in line with local legislation. • For further information on driving with ICDs, see local rules.
Occupation	• Most people with cardiomyopathy will be able to accomplish their normal jobs. The implications of heavily manual jobs that involve strenuous activity should be discussed with the appropriate specialist. • For some occupations, such as pilots, military personnel, and emergency services personnel, there are strict guidelines or rules on eligibility. • The social and financial implications of a diagnosis of cardiomyopathy should be included in the counselling of relatives before clinical or genetic screening.
Holidays and travel insurance	• Most asymptomatic or mildly symptomatic patients can fly safely. For further information on flying with ICD, see 'Fitness to fly for passengers with cardiovascular disease'.[1172] • Insurance companies may charge more for travel insurance. In some countries, patient support organizations can provide advice about obtaining reasonable insurance.
Life insurance	• The diagnosis of cardiomyopathy will result in difficulty obtaining life insurance or mortgages. Advice on the rules that apply in different countries should be provided to patients at diagnosis.
Pregnancy and childbirth	See *Section 8.2*
Education/schooling	• Teachers and other carers should be provided with advice and written information relevant to the care of children with cardiomyopathy. • In the absence of symptoms and risk factors, children should be allowed to perform low- to moderate-level aerobic physical activity, in accordance with advice from their paediatric cardiologist. Advice on high-intensity exercise in children should be guided by cardiomyopathy phenotype and the presence of symptoms and risk factors within a specialist paediatric cardiomyopathy setting. • Provision should be made for children with learning difficulties and other special needs. • Parents, teachers, and staff at sports facilities should be trained in CPR and in the use of AEDs.

AED, automated external defibrillator; CPR, cardio-pulmonary resuscitation; ICD, implantable cardioverter defibrillator; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

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Table 24 summarizes some of the key issues that should be discussed with patients, relatives, and carers. When appropriate (e.g. when considering pregnancy, see *Section 8.2*), patients should be referred to other specialist services.

11. Sex differences in cardiomyopathies

Sex differences in phenotypic expression and outcomes are well documented across cardiovascular medicine. Differences in clinical presentation, progression, and outcome in cardiomyopathies between females and males can be attributable to genetic and hormonal differences, but also to variations in management, access to healthcare, or response to specific therapies.[546,1173] Eliminating these variations represents a major unmet need in the care of cardiomyopathies.

Cardiomyopathies are typically inherited as an autosomal dominant trait. Therefore, the prevalence would be expected to be equal among the sexes. Women are consistently less represented than men in clinical studies across different cardiomyopathies (30–40%). Despite this, the difference may be explained by bias in interaction with healthcare facilities or by diagnosis criteria based on unadjusted cardiac imaging measurements; data from large pedigrees seem to support the hypothesis that there is a real delay in the age of phenotypic expression in female carriers (at least for HCM).[178,1174–1176]

Females with HCM are diagnosed later than males (8–13 years later), are more severely affected, more often have LVOTO, have more severe symptoms at baseline, and more commonly develop advanced heart failure during follow-up.[1177,1178] Women with LVOTO and indication for invasive procedures are often older and more symptomatic than males.[1179–1181] Females and males appear to show similar survival benefit from invasive SRT.[705,1181] Cardiomyopathy-related death has shown to be increased in middle-aged females with HCM compared with men and the general population; this is due to a higher rate of death from heart failure. No difference in SCD has been demonstrated in HCM regarding sex.[1182,1183]

Females with DCM may have a better response to therapy and seem to have a more favourable clinical course than males.[186,1184] Male sex has been reported to be consistently associated with an increased SCD rate in DCM (general cohorts and particular genotypes series),[186,541,872,878,1185–1187] and death from heart failure or transplant in general DCM cohorts.[1188,1189]

Male sex and sports have been traditionally identified as variables associated with an earlier phenotypic penetrance and a more severe disease expression in genetic carriers, and are independent predictors of malignant ventricular arrhythmic events in ARVC.[522,950,1190–1195] As in HCM, females with ARVC may have an increased risk of developing heart failure.[1193]

Reports on sex differences in familial or genetic RCM are scarce.[331,546] Compared with other types of cardiomyopathies, females seem to be as equally represented as males in RCM series.[331]

12. Comorbidities and cardiovascular risk factors in cardiomyopathies

12.1. Cardiovascular risk factors

The penetrance of the disease in genetic carriers of cardiomyopathy-associated variants is incomplete. Gene–environment

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interactions can explain part of the heterogeneity of the phenotypic expression of all cardiomyopathy phenotypes, although published data focus primarily on HCM, DCM, and ARVC.

12.2. Dilated cardiomyopathy

Individual genetic predisposition favours a dilated phenotype in the presence of trigger factors, such as inflammation, infection, toxic insults from alcohol or drugs, and tachyarrhythmias.

12.3. Hypertrophic cardiomyopathy

Hypertension and obesity have been associated with penetrance and phenotypic expression of HCM.[1196] Results from the EORP Cardiomyopathy/Mycarditis Registry showed that patients with HCM had a high prevalence of cardiovascular risk factors, comparable with data from the general population.[1196] Hypertension, diabetes, and obesity were associated with older age at presentation, a lower prevalence of family history of HCM and SCD, more symptoms, frequent AF, and worse LV diastolic function.[1197] Hypertension and obesity were also associated with higher provocable LVOT gradients and LVH.[1198]

12.4. Arrhythmogenic right ventricular cardiomyopathy

The role of intense exercise in disease expression and outcomes has been studied in HCM and DCM, but the impact has shown to be particularly relevant in ARVC (*Table 25*). Despite significant research, the pathophysiology of ARVC is complex and not well understood. The search for genetic or environmental triggers, such as viruses and immune response, has failed to identify actionable factors. The role of inflammation on the pathophysiology is thought to be key.[1199]

Recommendation Table 34 — Recommendation for management of cardiovascular risk factors in patients with cardiomyopathy

Recommendation	Class[a]	Level[b]
Identification and management of risk factors and concomitant diseases is recommended as an integral part of the management of cardiomyopathy patients.	I	C

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[a]Class of recommendation.
[b]Level of evidence.

13. Coronavirus disease (COVID-19) and cardiomyopathies

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection, known as COVID-19, is characterized by a high variability of clinical presentation and outcomes with an adverse association between underlying cardiac disease, including heart failure, and SARS-CoV-2-related mortality.[1200–1202] However, examination of SARS-CoV-2 infection in underlying causes of heart failure, particularly cardiomyopathies, has been limited.

Analyses of international registries on cardiomyopathies and SARS-CoV-2 from the pre-vaccine period have identified several markers of adverse outcomes.[1203] Prior history of heart failure and particular phenotypes (amyloidosis and DCM) were significantly associated with intensive care unit admission and death compared with HCM, ARVC, and the general population. For HCM, age, baseline functional class, LVOTO, and systolic impairment were independent predictors of death.[1204]

SARS-CoV-2 vaccination has been demonstrated to be safe in large population studies and reports on complications related to the vaccination in patients with cardiomyopathy are anecdotal. Given this, and the potential for worse outcomes in cardiomyopathy patients who contract COVID-19, vaccination is encouraged in all cardiomyopathy patients and, in particular, in those with signs or symptoms of heart failure.

14. Key messages

(1) Cardiomyopathies are more common than previously thought and they typically require nuanced management that may differ from the conventional approach to patients with arrhythmia or heart failure.
(2) Aetiology is fundamental to the management of patients with heart muscle disease and careful and systematic description of the morphological and functional phenotype is a crucial first step in the diagnostic pathway.
(3) An approach to nomenclature and diagnosis of cardiomyopathies that is based on the predominant phenotype at presentation is recommended.
(4) Patients with cardiomyopathy may seek medical attention due to symptoms onset (HF or arrhythmia related), incidental abnormal findings, or as a result of family screening following the diagnosis in a relative.

Table 25 Modulators of the phenotypic expression of cardiomyopathies

Condition	HCM	DCM	ARVC	Expression
Hypertension	+++	++	?	Hypertrophy, dilatation, dysfunction, AF
Diabetes	++	+	?	Hypertrophy, dysfunction, AF
Obesity	++	+	?	Hypertrophy, LVOTO, AF
Toxic	–	+++	?	Dilatation, dysfunction
Sports	+	+	+++	Dilatation, dysfunction, ventricular arrhythmia
Virus	–	++	+	Dilatation, dysfunction, ventricular arrhythmia
Pregnancy	–	++	–	Dilatation, dysfunction

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AF, atrial fibrillation; ARVC, arrhythmogenic right ventricular cardiomyopathy; DCM, dilated cardiomyopathy; HCM, hypertrophic cardiomyopathy; LVOTO, left ventricular outflow tract obstruction.
+, degree of positive association; –, absence of definitive association; ?, unknown association.

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(5) Multimodality imaging to characterize the cardiac phenotype (morphology and function)—including tissue characterization for non-ischaemic myocardial scar detection—is necessary, in combination with a detailed personal and family history, clinical examination, electrocardiography, and laboratory investigations.
(6) Imaging results should always be interpreted in the overall clinical context, including genetic testing results, rather than in isolation.
(7) Tissue characterization by CMR is of value in diagnosis, monitoring of disease progression and risk stratification in each of the main cardiomyopathy phenotypes.
(8) DPD/PYP/HMDP bone-tracer scintigraphy or SPECT represent the gold standard for the diagnosis of ATTR-related cardiac amyloidosis.
(9) The presence of non-ischaemic ventricular scar or fatty replacement on cardiac CMR and/or pathological examination, which can occur with or without ventricular dilatation and/or systolic dysfunction, can be the sole clue to the diagnosis of a cardiomyopathy and can have prognostic significance that varies with aetiology.
(10) The aim of this multiparametric and systemic approach is to generate a phenotype-based aetiological diagnosis, interpreting available data with a cardiomyopathy-oriented mindset that combines cardiological assessment with non-cardiac parameters.
(11) A multidisciplinary approach to patient care and appropriate transition of care from paediatric to adult cardiomyopathy services is needed.
(12) Genetic testing should be performed in patients with cardiomyopathy and may influence risk stratification and management.
(13) Genetic counselling, including pre- and post-test counselling, and psychological support are an essential aspect of the multidisciplinary care of patients with cardiomyopathy and their relatives.
(14) Paediatric cardiomyopathies largely represent part of the same clinical spectrum as those seen in older adolescents and adults, but infant-onset (in the first year of life) cardiomyopathies are often associated with severe phenotypes and a high rate of heart failure-related morbidity and mortality.
(15) Beyond the first year of life, genetic causes of childhood-onset cardiomyopathies are similar to those in adults.
(16) Symptom management, identification, and prevention of disease-related complications (including SCD, heart failure, and stroke) are the cornerstone of management of all cardiomyopathies.
(17) Cardiac myosin inhibitors (Mavacamten) should be considered in patients with HCM and LVOTO who remain symptomatic despite optimal medical therapy.
(18) Validated SCD risk-prediction tools (HCM Risk-SCD and HCM Risk-Kids) are the first step in sudden death prevention in patients with HCM.
(19) Additional risk markers may be of use in patients with low or intermediate risk, but there is a lack of robust data on the impact of these parameters on the personalized risk estimates generated by the risk-prediction tools.
(20) Pharmacological treatment of DCM patients does not differ from those recommended in chronic heart failure.
(21) SCD risk of DCM and NDLVC patients varies depending on the underlying cause and genetic subtype.
(22) CMR findings play an important role in guiding ICD implantation for patients with DCM and NDLVC.
(23) In DCM and NDLVC patients, ICD should be considered for certain genetic forms even if LVEF is >35%.
(24) It is of importance to define aetiology for a tailored management in patients with syndromic and metabolic cardiomyopathies (i.e. ERT/chaperone in lysosomal storage disease; tafamidis in ATTRwt, etc.).
(25) Pregnancy and the post-partum period are associated with increased cardiovascular risk in women with known cardiomyopathy.
(26) A multidisciplinary team should evaluate the patient with cardiomyopathy to assess the risk associated with pregnancy.
(27) Beta-blocker therapy on arrhythmic indication can safely be continued during pregnancy; safety data should be checked before initiation of new drugs in pregnancy.
(28) Healthy adults of all ages and individuals with known cardiac disease should exercise with moderate intensity, totalling at least 150 min per week.
(29) All patients with cardiomyopathy should have an individualized risk assessment for exercise prescription. Evaluation should be guided by three principles: (i) preventing life-threatening arrhythmias during exercise; (ii) symptom management to allow sports; and (iii) preventing sports-induced progression of the arrhythmogenic condition.
(30) Individuals who are genotype positive/phenotype negative or have a mild cardiomyopathy phenotype and absence of symptoms or any risk factors, may be able to participate in competitive sports. In some high-risk patients with HCM, ARVC, and NDLVC, high-intensity exercise and competitive sports should be discouraged.
(31) Patients with high-risk genotypes or associated factors for arrhythmic or heart failure complications or severe LVOTO should be referred for specialized investigations before undergoing elective NCS.
(32) Identification and management of risk factors and concomitant diseases is recommended as an integral part of the management of cardiomyopathy patients.

15. Gaps in evidence

Although there have been major advances in the genetics, diagnosis, and treatments of patients with cardiomyopathy over the last few years, there are a number of areas where robust evidence is still lacking and deserve to be addressed in future clinical research.

(1) Cardiomyopathy phenotypes.
(2) Epidemiology:
  (a) Prevalence of NDLVC phenotype (children and adults).
  (b) Systematic assessment of prevalence of cardiomyopathy phenotypes in childhood.
(3) Integrated patient management:
  (a) Embedding of telemedicine into cardiomyopathy networks.
(4) Patient pathway:
  (a) Laboratory tests:
    (i) Studies on novel 'omic' biomarkers (proteomics, metabolomics, and transcriptomics) are needed to assess their potential value for diagnostic and prognostic purposes in cardiomyopathies.
  (b) Multimodality imaging:
    (i) Advanced echocardiographic techniques, including speckle tracking deformation imaging, are promising but lack robust validation in the setting of cardiomyopathies.

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(ii) A universally accepted, standardized method for the quantification of myocardial fibrosis by CMR is lacking.
(iii) CMR scans may be performed in patients with compatible implantable devices, but the quality is limited by artefacts.
(iv) Artificial intelligence enhanced electrocardiography and imaging for cardiomyopathy evaluation has been proving a novel tool to dramatically improve diagnosis and prognosis; further studies are needed for routine introduction in clinical practice.
(v) Impact of CMR on screening in genotype-positive relatives of individuals with cardiomyopathy and in gene-elusive families.

(c) Genetics:
(i) Penetrance is poorly characterized for most pathogenic variants. This is true both for variants found through cascade screening of relatives of a patient with cardiomyopathy, and also for variants found in the wider population who may have clinical sequencing for another indication or may choose to have genome sequencing as a screening test.
(ii) The benefits, harm, and costs of screening of cardiomyopathy-associated genes in individuals without a personal or family history of cardiomyopathy is not known.

(d) General principles in management:
(i) Management of RV failure remains largely non-evidence-based.
(ii) Large-scale studies are required to guide ventricular arrhythmia management in patients with genetic cardiomyopathies.
(iii) Optimal rate control and AADs per subtype of cardiomyopathy.
(iv) The role of ICDs in patients with well tolerated VT.
(v) All risk calculators are developed using baseline data. Therefore, the utility of their application during follow-up visits of patients remains unclear and needs to be studied.
(vi) Risk prediction in childhood cardiomyopathies other than HCM remains empirical—multicentre approach required to understand and develop SCD risk models in childhood.
(vii) Lack of controlled studies on the effect of ablation in patients with AF and cardiomyopathy.
(viii) Models to predict AF recurrence have not been validated in cardiomyopathy patients.
(ix) Lack of randomized studies assessing the efficacy of cardiac sympathetic denervation for the prevention of VT/VF recurrences.

(e) Approach to paediatric cardiomyopathies:
(i) Lack of randomized studies or large registries addressing the benefit and optimal dosing of drug therapy in paediatric population.

(5) Hypertrophic cardiomyopathy:
(a) Epidemiology:
(i) Imaging and genotype studies suggest a population prevalence of up to 1 in 200 of the population. However, HER-based studies suggest a much lower number of 3–4/10 000. Further studies into the prevalence of clinically important diseases are necessary.

(b) Aetiology:
(i) Aetiology of gene-elusive disease.
(ii) The role of polygenic risk.
(iii) Interaction between comorbidity and disease outcomes.
(iv) Genetic and environmental determinants of disease expression in variant carriers.

(c) Symptom management:
(i) Optimal timing of LVOTO management and its impact on disease progression.
(ii) Prevention of AF and heart failure.

(d) Sudden death prevention:
(i) Impact of genetics (Mendelian and complex) on risk of disease-related outcomes.
(ii) Improved prediction models that reduce residual risk and prevent unnecessary ICD implantation.
(iii) Refinement of risk-prediction models to include serial data.
(iv) Role of LVOTO in risk prediction in children (apparent discrepancy compared with adults).

(e) New therapies:
(i) Clinical utility of myosin inhibitors, other small molecules, and emerging genetic therapies.

(6) Dilated cardiomyopathy:
(a) Genetic basis of familial DCM is still unknown in a high number of cases.
(b) Detailed data about the specific clinical course in diverse genetic and non-genetic DCM forms are not available.
(c) It is unknown if patients with DCM respond differently to pharmacological treatment according to underlying aetiology.
(d) Optimized SCD prevention strategy remains unsolved. There are not data from prospective clinical trials in modern cohorts with contemporary medical treatment. This gap is knowledge is particularly relevant for DCM patients with LVEF > 35%.
(e) Sport recommendations and utility of prophylactic pharmacological therapy to prevent DCM onset in genetic carriers.

(7) Non-dilated left ventricular cardiomyopathy:
(a) Prevalence of disease.
(b) Natural history and response to treatment.
(c) SCD prevention.
(d) Sports recommendations.

(8) Arrhythmogenic right ventricular cardiomyopathy:
(a) RCTs for therapies for the management of arrhythmias and heart failure are lacking.
(b) Studies on the effect of exercise remain largely retrospective.
(c) Studies on the incidence and prognostication of heart failure remain limited.
(d) Studies on the frequency and mode of clinical screening for asymptomatic family members are lacking.

(9) Restrictive cardiomyopathy:
(a) SCD prevention.

(10) Syndromic and metabolic cardiomyopathies:
(a) Lack of randomized trials or large observational cohort studies assessing the role of new target therapies addressing the RAS/MAPK pathway (i.e trametinib).
(b) There are few long-term outcome studies addressing ventricular remodelling in RAS-HCM.
(c) HCM Risk-Kids has not been validated in paediatric patients with RAS-HCM. Data regarding SCD risk stratification are lacking, although candidate risk factors have been identified.

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(d) Lack of studies addressing the optimal timing to start ERT in adolescents and adults with late-onset Pompe disease.
(e) Lack of standardized protocols to treat cross-reactive immunologic material-negative patients.
(f) Lack of standardization of clinical endpoints in ERT/chaperone therapy trials.
(g) Lack of head-to-head comparisons between agalsidase alpha and beta.
(h) Optimal time to begin treatment in asymptomatic female patients with non-classic disease.

(11) Amyloid:
(a) Further studies are needed to assess the efficacy and safety of tafamidis in NYHA class III patients.
(b) SCD risk stratification and indications for ICD implantation should be carefully defined, taking into account the estimated life expectancy, competitive non-cardiovascular mortality, and the high rate of pulseless electrical activity.
(c) The need for drug therapy in patients with cardiac amyloidosis and subclinical cardiac involvement (i.e asymptomatic patients, positive scintigraphy with negative ECHO) has not been clearly defined.

(12) Sports:
(a) 'Return to play' for patients with low-risk cardiomyopathies (and how to define low risk in relation to exercise).
(b) SCD risk and exercise recommendations in phenotype-negative gene carriers.
(c) Role of exercise in disease expression and progression.
(d) Large, adequately powered randomized prospective studies are necessary to provide evidence-based recommendations for optimal exercise prescription without compromising safety.

(13) Reproductive issues:
(a) Several cardiomyopathies lack specific outcome data regarding pregnancy.
(b) There is a lack of randomized trials on the use of AADs, heart failure drugs, and interventions during pregnancy.

(14) Non-cardiac interventions:
(a) There is a lack specific outcome data regarding risks of non-cardiac interventions.

(15) Management of cardiovascular risk factors in patients with cardiomyopathies:
(a) There is a lack of data on the impact of comorbidities on penetrance, severity, and outcome of cardiomyopathies.

16. 'What to do' and 'What not to do' messages from the Guidelines

Recommendations	Class[a]	Level[b]
Recommendations for the provision of service of multidisciplinary cardiomyopathy teams		
It is recommended that all patients with cardiomyopathy and their relatives have access to multidisciplinary teams with expertise in the diagnosis and management of cardiomyopathies.	I	C
Timely and adequate preparation for transition of care from paediatric to adult services, including joint consultations, is recommended in all adolescents with cardiomyopathy.	I	C
Recommendations for diagnostic work-up in cardiomyopathies		
It is recommended that all patients with suspected or established cardiomyopathy undergo systematic evaluation using a multiparametric approach that includes clinical evaluation, pedigree analysis, ECG, Holter monitoring, laboratory tests, and multimodality imaging.	I	C
It is recommended that all patients with suspected cardiomyopathy undergo evaluation of family history and that a three- to four-generation family tree is created to aid in diagnosis, provide clues to underlying aetiology, determine inheritance pattern, and identify at-risk relatives.	I	C
Recommendations for laboratory tests in the diagnosis of cardiomyopathies		
Routine (first-level) laboratory tests are recommended in all patients with suspected or confirmed cardiomyopathy to evaluate aetiology, assess disease severity, and aid in detection of extracardiac manifestations and assessment of secondary organ dysfunction.	I	C
Recommendation for echocardiographic evaluation in patients with cardiomyopathy		
A comprehensive evaluation of cardiac dimensions and LV and RV systolic (global and regional) and LV diastolic function is recommended in all patients with cardiomyopathy at initial evaluation, and during follow-up, to monitor disease progression and aid risk stratification and management.	I	B
Recommendations for cardiac magnetic resonance indication in patients with cardiomyopathy		
Contrast-enhanced CMR is recommended in patients with cardiomyopathy at initial evaluation.	I	B
Recommendations for computed tomography and nuclear imaging		
DPD/PYP/HMDP bone-tracer scintigraphy is recommended in patients with suspected ATTR-related cardiac amyloidosis to aid diagnosis.	I	B
Recommendations for genetic counselling and testing in cardiomyopathies		
Genetic counselling		
Genetic counselling, provided by an appropriately trained healthcare professional and including genetic education to inform decision-making and psychosocial support, is recommended for families with an inherited or suspected inherited cardiomyopathy, regardless of whether genetic testing is being considered.	I	B

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It is recommended that genetic testing for cardiomyopathy is performed with access to a multidisciplinary team, including those with expertise in genetic testing methodology, sequence variant interpretation, and clinical application of genetic testing, typically in a specialized cardiomyopathy service or in a network model with access to equivalent expertise.	I	B
Pre- and post-test genetic counselling is recommended in all individuals undergoing genetic testing for cardiomyopathy.	I	B
If pre-natal diagnostic testing is to be pursued by the family, it is recommended that this is performed early in pregnancy, to allow decisions regarding continuation or co-ordination of pregnancy to be made.	I	C
Index patients		
Genetic testing is recommended in patients fulfilling diagnostic criteria for cardiomyopathy in cases where it enables diagnosis, prognostication, therapeutic stratification, or reproductive management of the patient, or where it enables cascade genetic evaluation of their relatives who would otherwise be enrolled into long-term surveillance.	I	B
Genetic testing is recommended for a deceased individual identified to have cardiomyopathy at *post-mortem* if a genetic diagnosis would facilitate management of surviving relatives.	I	C
Family members		
It is recommended that cascade genetic testing, with pre- and post-test counselling, is offered to adult at-risk relatives if a confident genetic diagnosis (i.e. a P/LP variant) has been established in an individual with cardiomyopathy in the family (starting with first-degree relatives if available, and cascading out sequentially).	I	B
Diagnostic genetic testing is not recommended in a phenotype-negative relative of a patient with cardiomyopathy in the absence of a confident genetic diagnosis (i.e. a P/LP variant) in the family.	III	C
Recommendations for cardiac transplantation in patients with cardiomyopathy		
Orthotopic cardiac transplantation is recommended for eligible cardiomyopathy patients with advanced heart failure (NYHA class III–IV) or intractable ventricular arrhythmia refractory to medical/invasive/device therapy, and who do not have absolute contraindications.	I	C
Recommendations for management of atrial fibrillation and atrial flutter in patients with cardiomyopathy		
Anticoagulation		
Oral anticoagulation in order to reduce the risk of stroke and thromboembolic events is recommended in all patients with HCM or cardiac amyloidosis and AF or atrial flutter (unless contraindicated).	I	B
Oral anticoagulation to reduce the risk of stroke and thrombo-embolic events is recommended in patients with DCM, NDLVC, or ARVC, and AF or atrial flutter with a CHA_2DS_2-VASc score ≥ 2 in men or ≥ 3 in women.	I	B
Control of symptoms and heart failure		
Atrial fibrillation catheter ablation is recommended for rhythm control after one failed or intolerant class I or III AAD to improve symptoms of AF recurrences in patients with paroxysmal or persistent AF and cardiomyopathy.	I	B
Atrial fibrillation catheter ablation is recommended to reverse LV dysfunction in AF patients with cardiomyopathy when a tachycardia-induced component is highly probable, independent of their symptom status.	I	B
Comorbidities and associated risk factor management		
Modification of an unhealthy lifestyle and targeted therapy of intercurrent conditions is recommended to reduce AF burden and symptom severity in patients with cardiomyopathy.	I	B
Recommendations for implantable cardioverter defibrillator in patients with cardiomyopathy		
General recommendations		
Implantation of a cardioverter defibrillator is only recommended in patients who have an expectation of good quality survival >1 year.	I	C
It is recommended that ICD implantation be guided by shared decision-making that: • is evidence-based; • considers a person's individual preferences, beliefs, circumstances, and values; and • ensures that the person understands the benefits, harm, and possible consequences of different treatment options.	I	C
It is recommended that prior to ICD implantation, patients are counselled on the risk of inappropriate shocks, implant complications, and the social, occupational, and driving implications of the device.	I	C
It is not recommended to implant an ICD in patients with incessant ventricular arrhythmias until the ventricular arrhythmia is controlled.	III	C
Secondary prevention		
Implantation of an ICD is recommended:		
• in patients with HCM, DCM, and ARVC who have survived a cardiac arrest due to VT or VF, or who have spontaneous sustained ventricular arrhythmia causing syncope or haemodynamic compromise in the absence of reversible causes.	I	B
• in patients with NDLVC and RCM who have survived a cardiac arrest due to VT or VF, or who have spontaneous sustained ventricular arrhythmia causing syncope or haemodynamic compromise in the absence of reversible causes.	I	C

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Primary prevention		
Comprehensive SCD risk stratification is recommended in all cardiomyopathy patients who have not suffered a previous cardiac arrest/sustained ventricular arrhythmia at initial evaluation and at 1–2 year intervals, or whenever there is a change in clinical status.	I	C
The use of validated SCD algorithms/scores as aids to the shared decision-making when offering ICD implantation, where available is recommended in patients with HCM.	I	B
Choice of ICD		
When an ICD is indicated, it is recommended to evaluate whether the patient could benefit from CRT.	I	A
Recommendations for routine follow-up of patients with cardiomyopathy		
It is recommended that all clinically stable patients with cardiomyopathy undergo routine follow-up using a multiparametric approach that includes ECG and echocardiography every 1–2 years.	I	C
Clinical evaluation with ECG and multimodality imaging is recommended in patients with cardiomyopathy whenever there is a substantial or unexpected change in symptoms.	I	C
Recommendations for family screening and follow-up evaluation of relatives		
Following cascade genetic testing, clinical evaluation using a multiparametric approach that includes ECG and cardiac imaging and long-term follow-up is recommended in first-degree relatives who have the same disease-causing variant as the proband.	I	B
Following cascade genetic testing, it is recommended that first-degree relatives without a phenotype who do not have the same disease-causing variant as the proband are discharged from further follow-up but advised to seek re-assessment if they develop symptoms or when new clinically relevant data emerge in the family.	I	C
It is recommended that when no P/LP variant is identified in the proband or genetic testing is not performed, an initial clinical evaluation using a multiparametric approach that includes ECG and cardiac imaging is performed in first-degree relatives.	I	C
Recommendations for psychological support in patients and family members with cardiomyopathies		
It is recommended that psychological support by an appropriately trained health professional be offered to all individuals who have experienced the premature sudden cardiac death of a family member with cardiomyopathy.	I	B
It is recommended that psychological support by an appropriately trained health professional be offered to all individuals with an inherited cardiomyopathy who receive an implantable cardioverter defibrillator.	I	B
Recommendation for evaluation of left ventricular outflow tract obstruction		
In all patients with HCM, at initial evaluation, transthoracic 2D and Doppler echocardiography are recommended, at rest and during Valsalva manoeuvre in the sitting and semi-supine positions—and then on standing if no gradient is provoked—to detect LVOTO.	I	B
In symptomatic patients with HCM and a resting or provoked peak instantaneous LV outflow tract gradient <50 mmHg, 2D and Doppler echocardiography during exercise in the standing, sitting (when possible), or semi-supine position are recommended to detect provocable LVOTO and exercise-induced mitral regurgitation.	I	B
Recommendations for medical treatment of left ventricular outflow tract obstruction		
Non-vasodilating beta-blockers, titrated to maximum tolerated dose, are recommended as first-line therapy to improve symptoms in patients with resting or provoked LVOTO.	I	B
Verapamil or diltiazem, titrated to maximum tolerated dose, are recommended to improve symptoms in symptomatic patients with resting or provoked LVOTO who are intolerant or have contraindications to beta-blockers.	I	B
Disopyramide, titrated to maximum tolerated dose, is recommended in addition to a beta-blocker (or, if this is not possible, with verapamil or diltiazem) to improve symptoms in patients with resting or provoked LVOTO.	I	B
Recommendations for septal reduction therapy		
It is recommended that SRT be performed by experienced operators working as part of a multidisciplinary team expert in the management of HCM.	I	C
SRT to improve symptoms is recommended in patients with a resting or maximum provoked LVOT gradient of ≥50 mmHg who are in NYHA/Ross functional class III–IV, despite maximum tolerated medical therapy.	I	B
Septal myectomy, rather than ASA, is recommended in children with an indication for SRT, as well as in adult patients with an indication for SRT and other lesions requiring surgical intervention (e.g. mitral valve abnormalities).	I	C
Additional recommendations for prevention of sudden cardiac death in patients with hypertrophic cardiomyopathy		
Secondary prevention		
Implantation of an ICD is recommended in patients who have survived a cardiac arrest due to VT or VF, or who have spontaneous sustained VT with haemodynamic compromise.	I	B

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Primary prevention		
The HCM Risk-SCD calculator is recommended as a method of estimating risk of sudden death at 5 years in patients aged ≥16 years for primary prevention.	I	B
Validated paediatric-specific risk-prediction models (e.g. HCM Risk-Kids) are recommended as a method of estimating risk of sudden death at 5 years in patients aged <16 years for primary prevention.	I	B
It is recommended that the 5-year risk of SCD be assessed at first evaluation and re-evaluated at 1–2 year intervals or whenever there is a change in clinical status.	I	B
Recommendations for an implantable cardioverter defibrillator in patients with dilated cardiomyopathy		
Secondary prevention		
An ICD is recommended to reduce the risk of sudden death and all-cause mortality in patients with DCM who have survived a cardiac arrest or have recovered from a ventricular arrhythmia causing haemodynamic instability.	I	B
Recommendation for resting and ambulatory electrocardiogram monitoring in patients with non-dilated left ventricular cardiomyopathy		
Ambulatory ECG monitoring is recommended in patients with NDLVC annually or when there is a change in clinical status, to aid in management and risk stratification.	I	C
Recommendations for an implantable cardioverter defibrillator in patients with non-dilated left ventricular cardiomyopathy		
An ICD is recommended to reduce the risk of sudden death and all-cause mortality in patients with NDLVC who have survived a cardiac arrest or have recovered from a ventricular arrhythmia causing haemodynamic instability.	I	C
Recommendation for resting and ambulatory electrocardiogram monitoring in patients with arrhythmogenic right ventricular cardiomyopathy		
Annual ambulatory ECG monitoring is recommended in patients with ARVC to aid in diagnosis, management, and risk stratification.	I	C
Recommendations for the antiarrhythmic management of patients with arrhythmogenic right ventricular cardiomyopathy		
Beta-blocker therapy is recommended in ARVC patients with VE, NSVT, and VT.	I	C
Recommendations for sudden cardiac death prevention in patients with arrhythmogenic right ventricular cardiomyopathy		
Secondary prevention		
An ICD is recommended to reduce the risk of sudden death and all-cause mortality in patients with ARVC who have survived a cardiac arrest or have recovered from a ventricular arrhythmia causing haemodynamic instability.	I	A
Recommendations for the management of patients with restrictive cardiomyopathy		
It is recommended that multimodality imaging be used to differentiate RCM from HCM or DCM with restrictive physiology.	I	C
It is recommended that baseline cardiac and non-cardiac investigations are performed to assess involvement of the neuromuscular system or other syndromic disorders.	I	C
Cardiac catheterization is recommended in all children with RCM to measure pulmonary artery pressures and PVR at diagnosis and at 6–12 monthly intervals to assess change in PVR.	I	B
ICD implantation is recommended to reduce the risk of sudden death and all-cause mortality in patients with RCM who have survived a cardiac arrest or have recovered from a ventricular arrhythmia causing haemodynamic instability.	I	C
Exercise recommendations for cardiomyopathy patients		
All cardiomyopathies		
Regular low- to moderate-intensity exercise is recommended in all able individuals with cardiomyopathy.	I	C
An individualized risk assessment for exercise prescription is recommended in all patients with cardiomyopathy.	I	C
HCM		
High-intensity exercise, including competitive sport, is not recommended in high-risk individuals and in individuals with left ventricular outflow tract obstruction and exercise-induced complex ventricular arrhythmias.	III	C
ARVC		
Moderate- and/or high-intensity exercise, including competitive sport, is not recommended in individuals with ARVC.	III	B
DCM and NDLVC		
High-intensity exercise, including competitive sport, is not recommended in symptomatic individuals, those with a left ventricular ejection fraction ≤40%, exercise-induced arrhythmias, or pathogenic variants in *LMNA* or *TMEM43*.	III	C
Recommendations for reproductive issues in patients with cardiomyopathy		
Pre-pregnancy risk assessment and counselling are recommended in all women using the mWHO classification of maternal risk.	I	C

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Counselling on safe and effective contraception is recommended in all women of fertile age and their partners.	I	C
Counselling on the risk of disease inheritance is recommended for all men and women before conception.	I	C
Vaginal delivery is recommended in most women with cardiomyopathies, unless there are obstetric indications for caesarean section, severe heart failure (EF <30% or NYHA class III–IV), or severe outflow tract obstructions, or in women presenting in labour on oral anticoagulants.	I	C
It is recommended that medication be carefully reviewed for safety in advance of pregnancy and adjusted according to tolerability in pregnancy.	I	C
Therapeutic anticoagulation with LMWH or VKAs according to the stage of pregnancy is recommended for patients with AF.	I	C
Recommendations for non-cardiac surgery in patients with cardiomyopathy		
Peri-operative ECG monitoring is recommended for all patients with cardiomyopathy undergoing surgery.	I	C
In patients with cardiomyopathy and suspected or known HF scheduled for intermediate or high-risk NCS, it is recommended to re-evaluate LV function with echocardiography (assessing LVOTO in HCM patients) and measurement of NT-proBNP/BNP levels, unless this has recently been performed.	I	B
It is recommended that cardiomyopathy patients with high-risk genotypes or associated factors for arrhythmic or heart failure complications or severe LVOTO be referred for additional specialized investigations to a cardiomyopathy unit before undergoing elective NCS.	I	C
In patients aged <65 years with a first-degree relative with a cardiomyopathy, it is recommended to perform an ECG and TTE before NCS, regardless of symptoms.	I	C
Recommendation for management of cardiovascular risk factors in patients with cardiomyopathy		
Identification and management of risk factors and concomitant diseases is recommended as an integral part of the management of cardiomyopathy patients.	I	C

2D, two-dimensional; AAD, antiarrhythmic drug; AF, atrial fibrillation; ARVC, arrhythmogenic right ventricular cardiomyopathy; ASA, alcohol septal ablation; ATTR, transthyretin amyloidosis; BNP, brain natriuretic peptide; CHA_2DS_2-VASc, congestive heart failure or left ventricular dysfunction, hypertension, age ≥75 (doubled), diabetes, stroke (doubled)-vascular disease, age 65–74, sex category (female) (score); CMR, cardiac magnetic resonance; CRT, cardiac resynchronization therapy; DCM, dilated cardiomyopathy; DPD, 3,3-diphosphono-1,2-propanodicarboxylic acid; ECG, electrocardiogram; EF, ejection fraction; HCM, hypertrophic cardiomyopathy; HMDP, hydroxymethylene diphosphonate; ICD, implantable cardioverter defibrillator; LMWH, low-molecular-weight heparin; LV, left ventricular; LVOT, left ventricular outflow tract; LVOTO, left ventricular outflow tract obstruction; mWHO, modified World Health Organization; NCS, non-cardiac surgery; NDLVC, non-dilated left ventricular cardiomyopathy; NSVT, non-sustained ventricular tachycardia; NT-proBNP, N-terminal pro-brain natriuretic peptide; NYHA, New York Heart Association; P/LP, pathogenic/likely pathogenic; PVR, pulmonary vascular resistance; PYP, pyrophosphate; RCM, restrictive cardiomyopathy; RV, right ventricular; SCD, sudden cardiac death; SRT, septal reduction therapy; TTE, transthorasic echocardiogram; VE, ventricular ectopic beats; VF, ventricular fibrillation; VKA, vitamin K antagonist; VT, ventricular tachycardia.
[a]Class of recommendation.
[b]Level of evidence.

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17. Supplementary data

Supplementary material is available at *European Heart Journal* online.

18. Data availability statement

No new data were generated or analysed in support of this research.

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20. Appendix

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21. Acknowledgements

The Task Force Chairs thank Sebastian Onciul for providing cardiac magnetic resonance images in Figure 7.

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